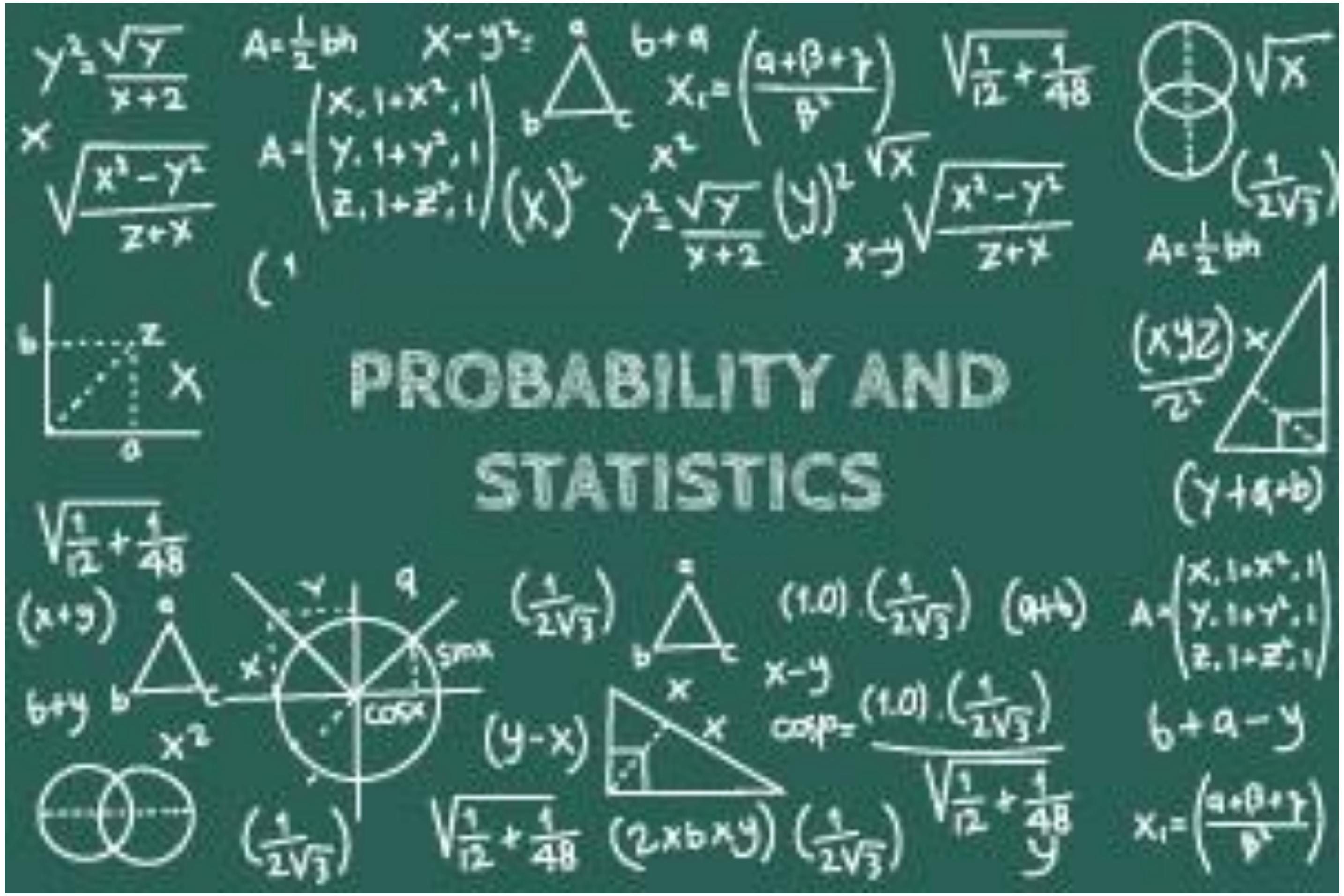
Lec # 08
ES111

Practice

Book Reading: Sec. 2.6

@M. Tahir Naseem



Discussion Plan

1. Summary

2. Practice

(Sec. 2.1-2.6)

1. Addition Rule

1. Addition Rule $\longrightarrow$ $\boxed{P(A \cup B) = P(A) + P(B) - P(A \cap B)}$

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2. Conditional Probability $\rightarrow$ $P(A/B) = \dfrac{P(A \cap B)}{P(B)}$; $P(\ /\ ) = \dfrac{P(\ \cap\ )}{P(\ )}$

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6. $P(A) = 1 - P(A')$

5. Independent Events $\rightarrow$ $\boxed{P(A) = P(A/B) \; ; \; P(B) = P(B/A)}$

One bag contains 4 red balls and 3 black balls, and a second bag contains 3 red balls and 5 black balls. One ball is drawn from the first bag and placed unseen in the second bag. What is the probability that a ball now drawn from the second bag is black?

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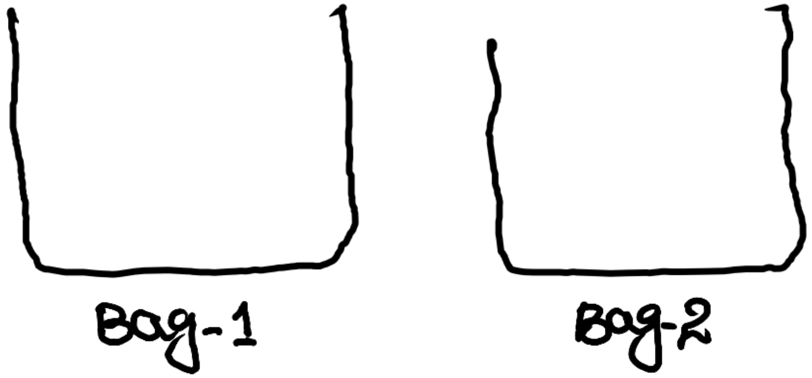

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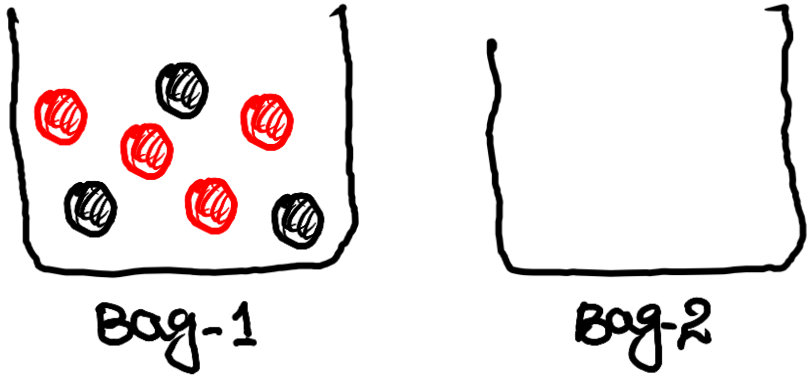

$R_1 = 4$

$B_1 = 3$

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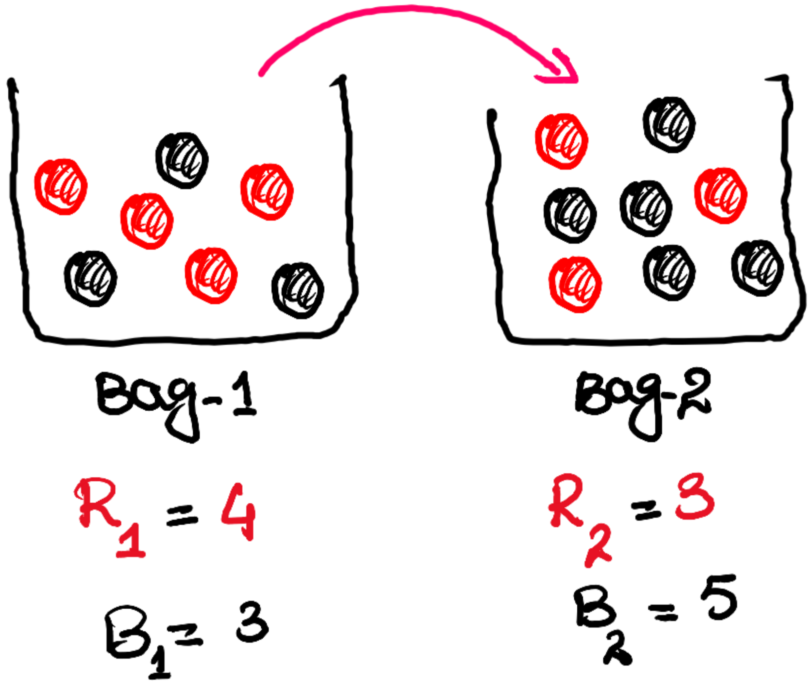

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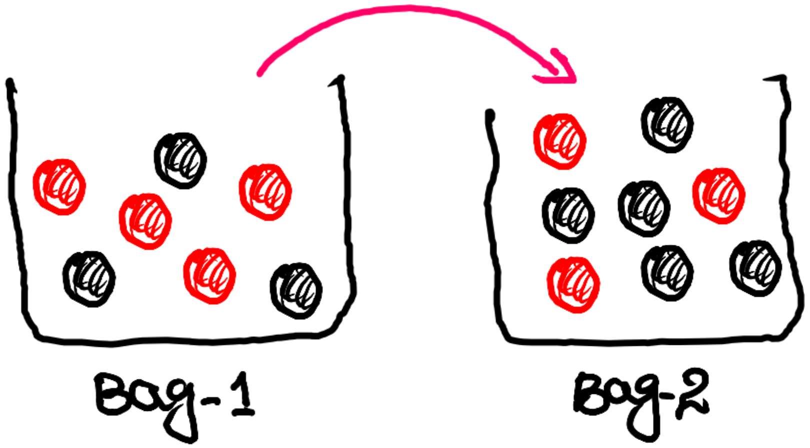
$P[(B_1 \cap B_2)$



$R_1 = 4$ $\quad$ $R_2 = 3$

$B_1 = 3$ $\quad$ $B_2 = 5$

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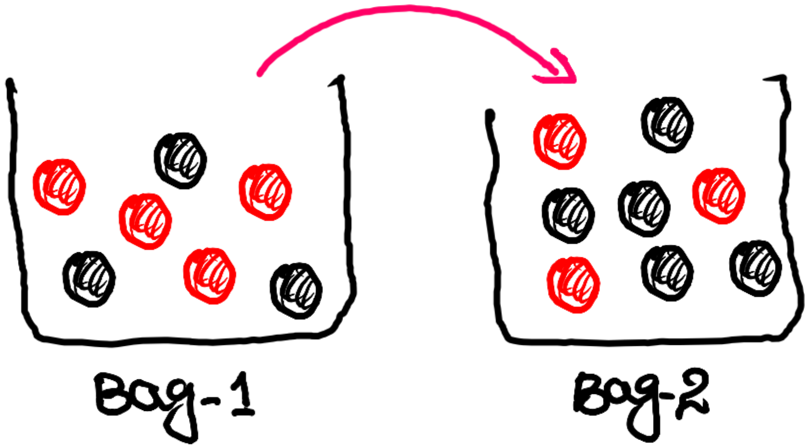
$P[(B_1 \cap B_2) \text{ or } (R_1 \cap B_2)]$



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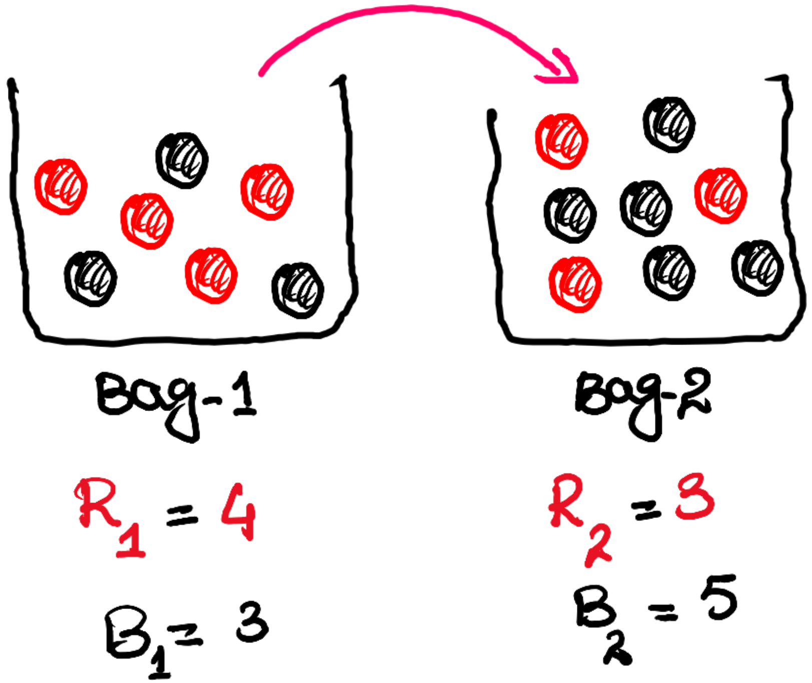
$$P[(B_1 \cap B_2) \text{ or } (R_1 \cap B_2)] =$$

$$P[(B_1 \cap B_2) \cup (R_1 \cap B_2)] =$$



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$$P(A \cup B) =$$

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$$= \frac{3}{7}$$

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$$= \frac{3}{7} \times \frac{6}{9}$$

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$$= P(B_1)\,P(B_2/B_1) + P(R_1)\,P(B_2/R_1)$$

$$= \frac{3}{7} \times \frac{6}{9} + \frac{4}{7}$$

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$= P(B_1)\, P(B_2/B_1) + P(R_1)\, P(B_2/R_1)$

$= \frac{3}{7} \times \frac{6}{9} + \frac{4}{7} \times \frac{5}{9}$

The product rule

$$\boxed{P(A \cap B) = P(B)\, P(A \mid B)}$$

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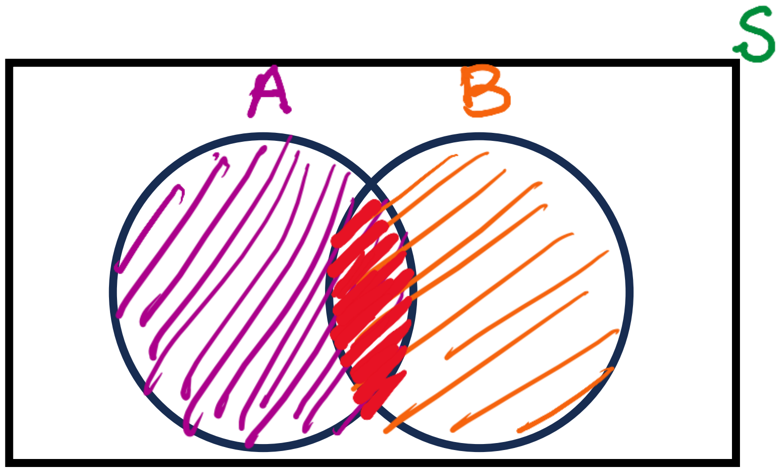

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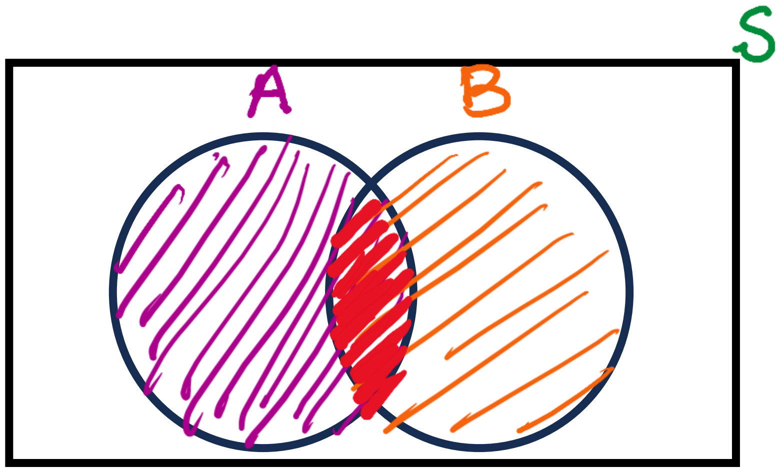

For Independent Events

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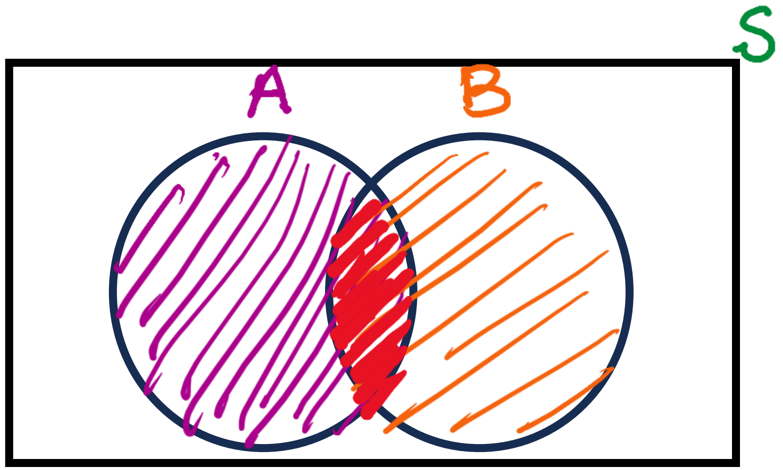

For Independent Events

$$P(A/B) = P(A)$$

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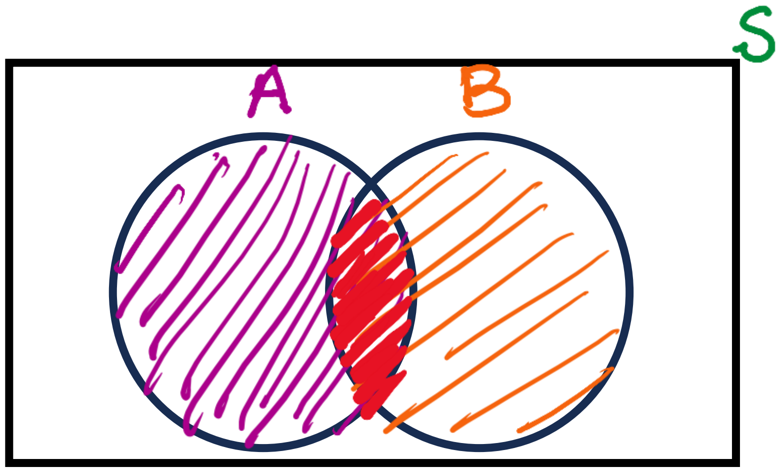

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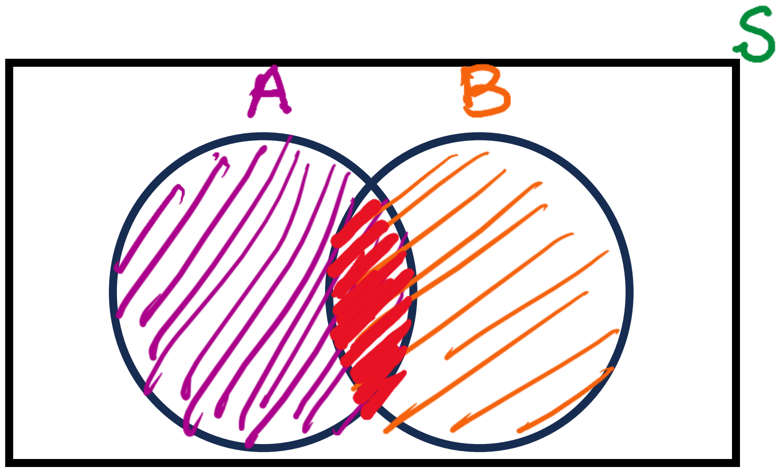

For Independent Events

$$P(A/B) = P(A)$$

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A small town has one fire engine and one ambulance available for emergencies. The probability that the fire engine is available when needed is 0.98, and the probability that the ambulance is available when called is 0.92. In the event of an injury resulting from a burning building, find the probability that both the ambulance and the fire engine will be available, assuming they operate independently.

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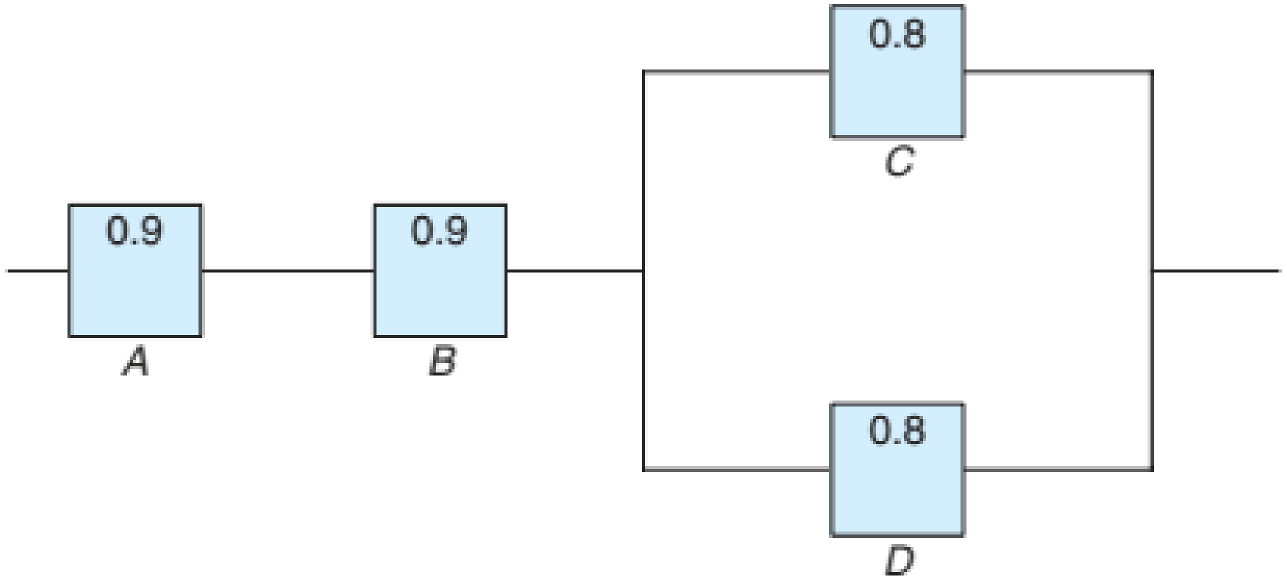
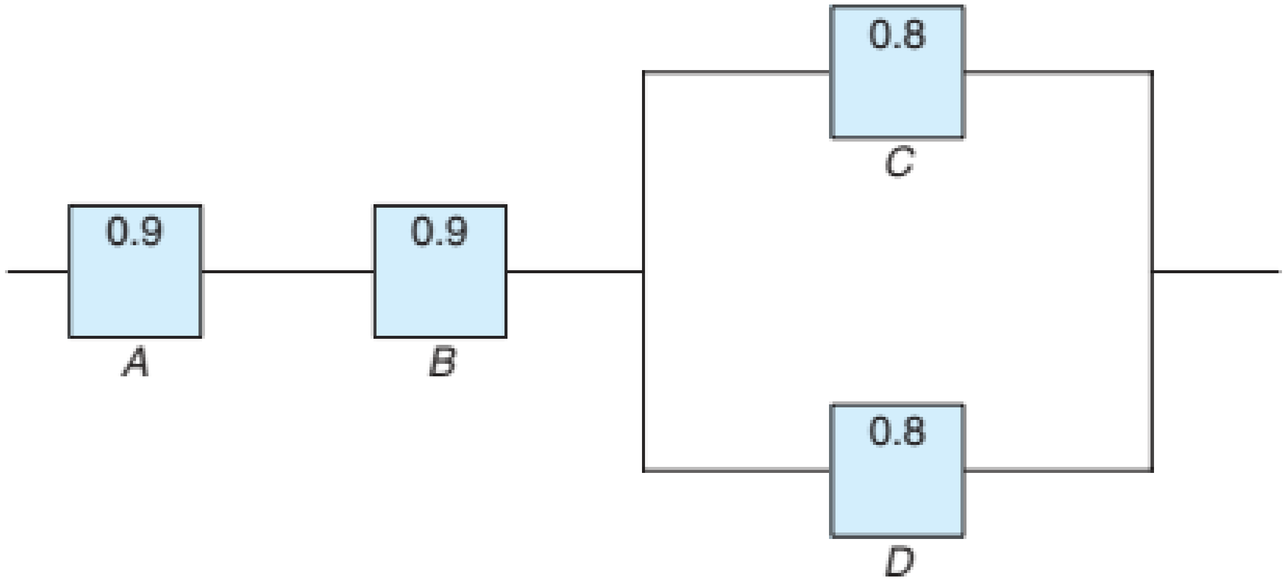
$$P(A \cap B) = P(A)\,P(B|A) = P(A)\,P(B)$$

$$= 0.98 \times 0.92 \quad \underline{\underline{\text{Ans}}}$$

An electrical system consists of four components as illustrated in Figure. The system works if components A and B work and either of the components C or D works. The reliability (probability of working) of each component is also shown in Figure. Find the probability that (a) the entire system works and (b) the component C does not work, given that the entire system works. Assume that the four components work independently.



The system works if components A and B work **and either of the components C or D works**. The probability of working of each component is shown in Figure. **Find the probability that (a) the entire system works.**



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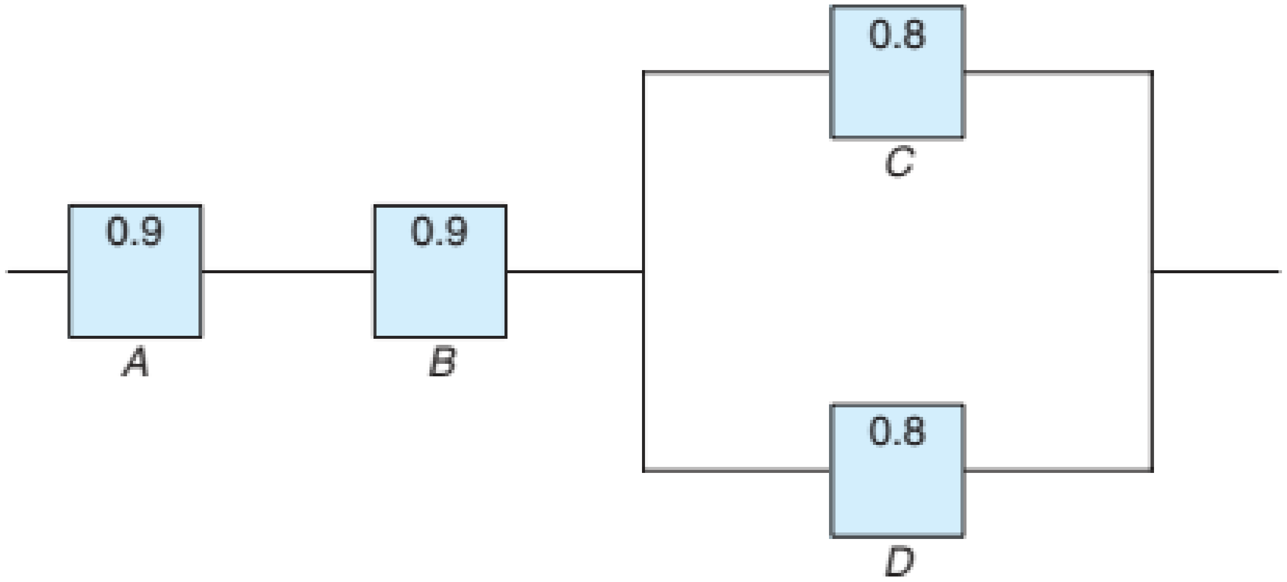
$P(A \cap B$



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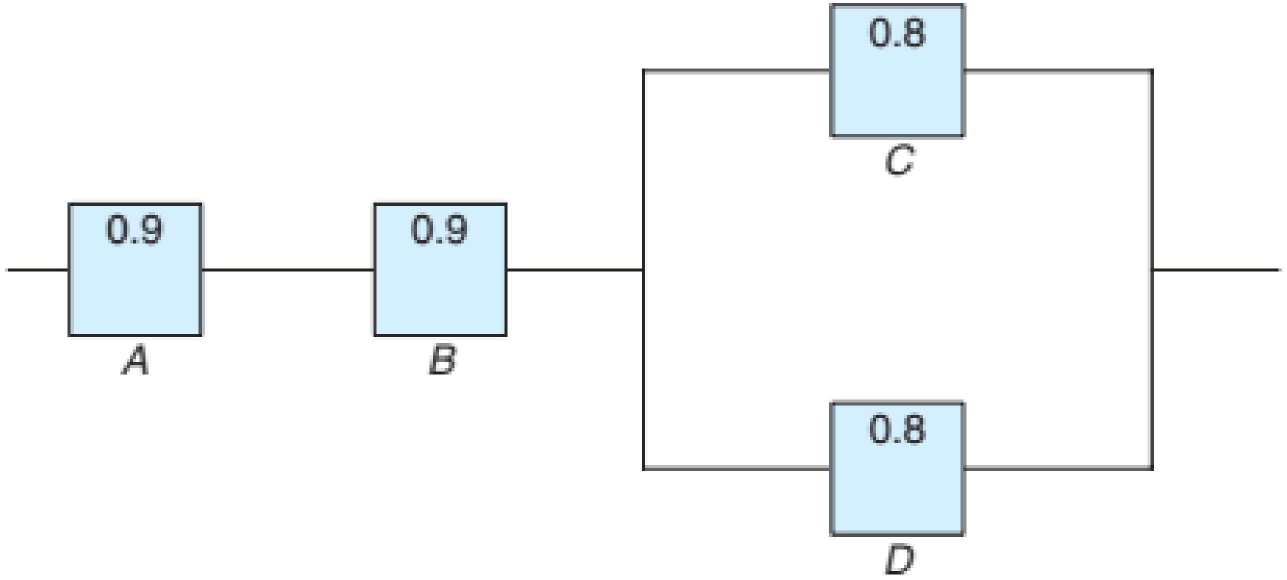

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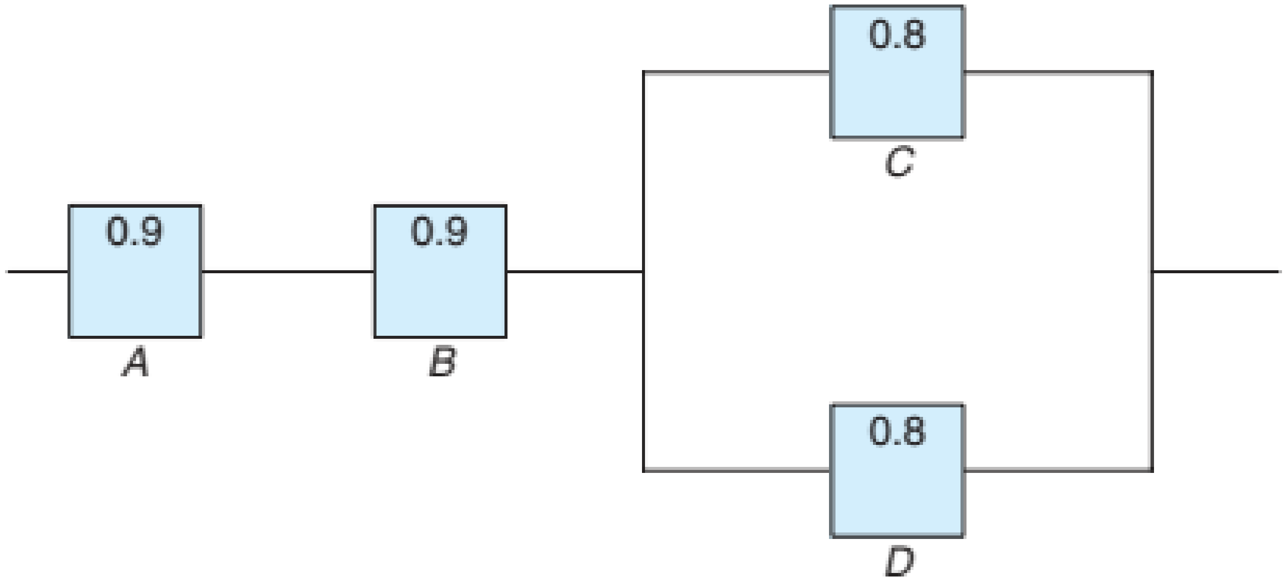
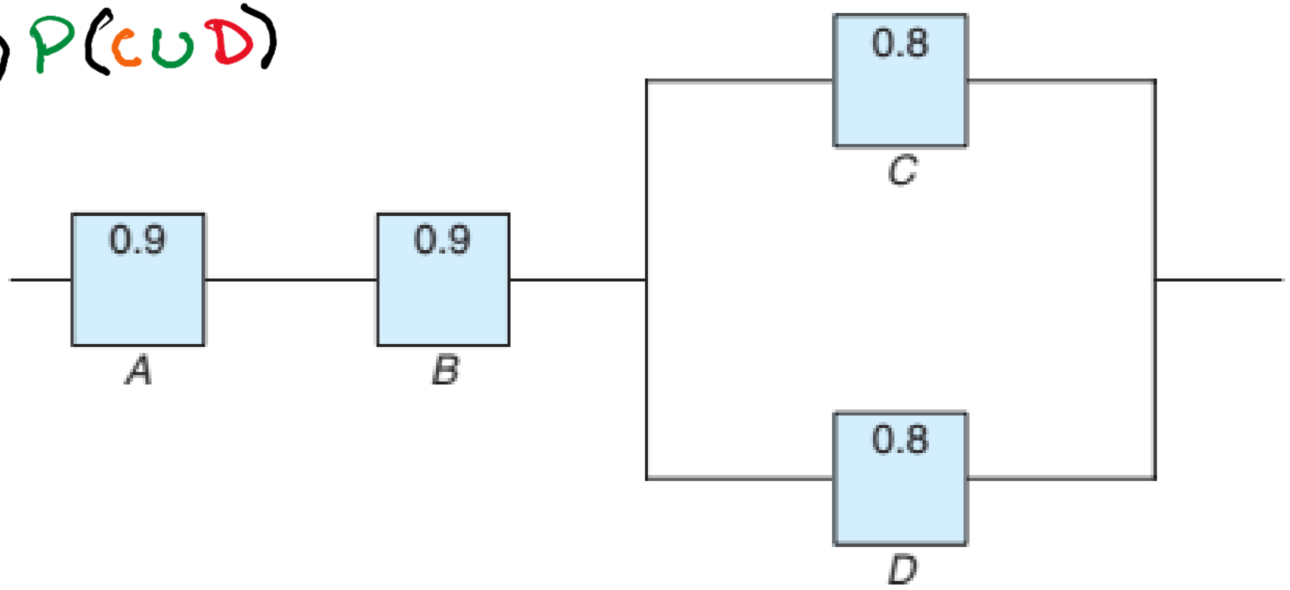
$P(A \cap B \cap (C \cup D)) =$



The system works if components A and B work **and** either **of the** components C **or** D works. The probability of working of each component is shown in Figure. **Find the probability that (a) the entire system works.**

$P(A \cap B \cap (C \cup D)) = P(A)P(B)P(C \cup D)$



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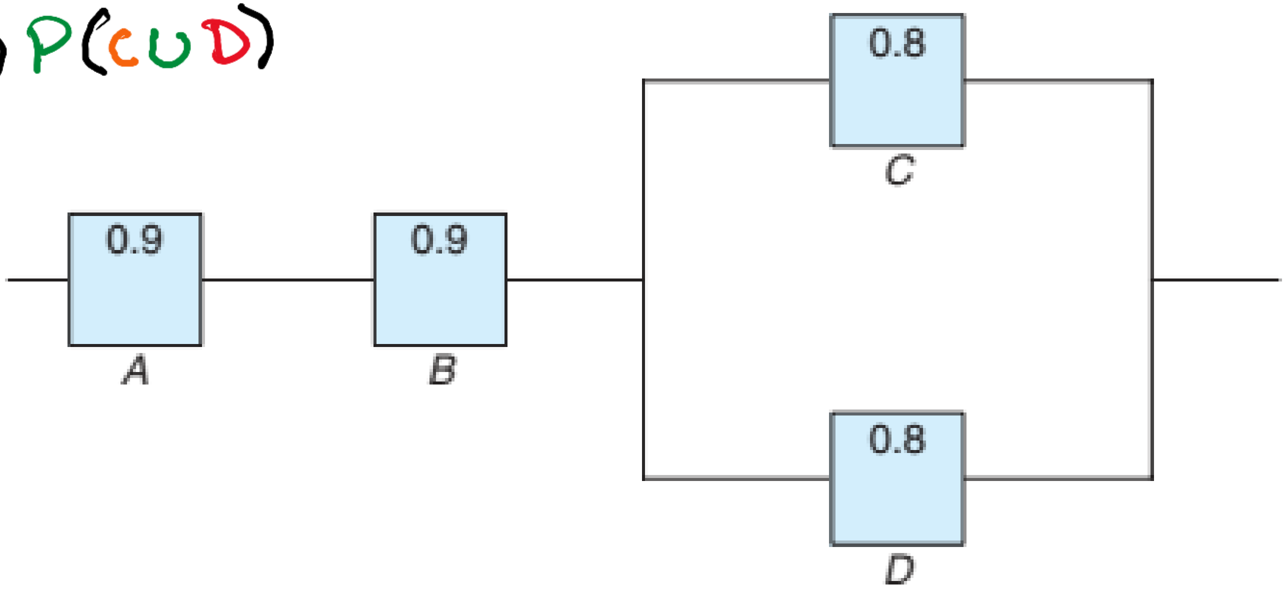

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$$P(A \cap B \cap (C \cup D)) = P(A)P(B)P(C \cup D)$$

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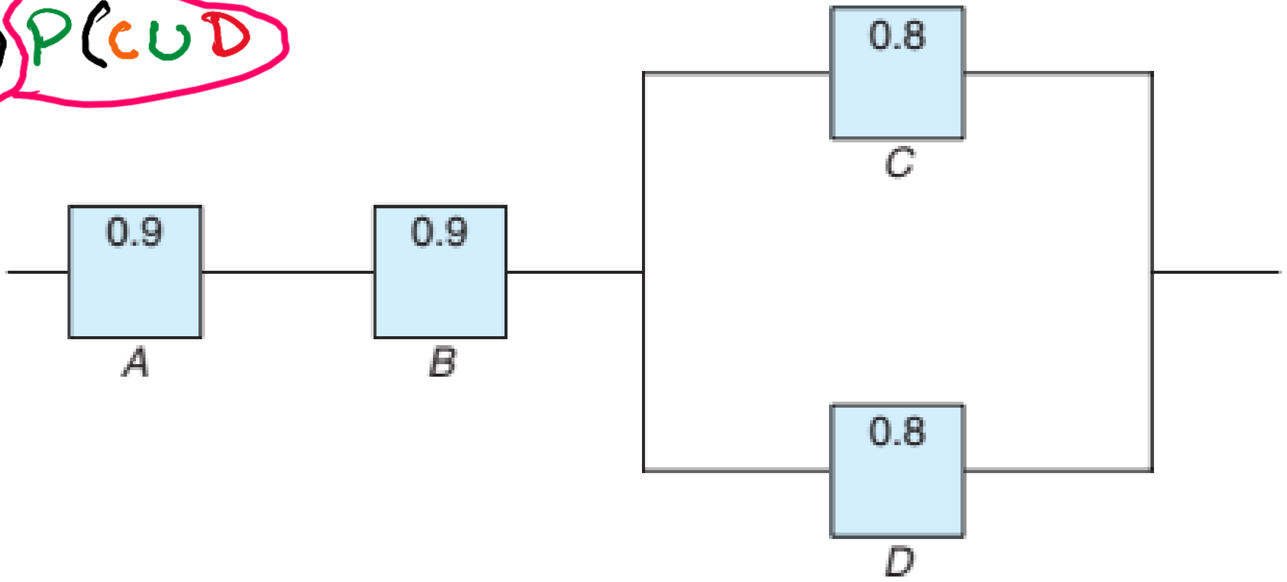

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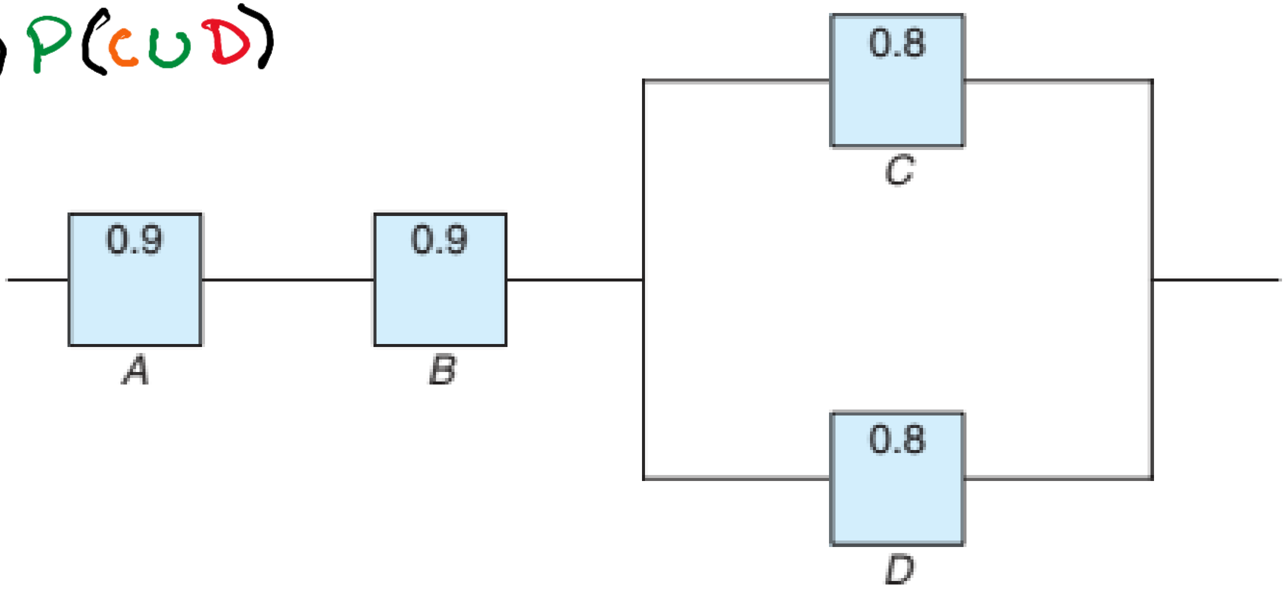

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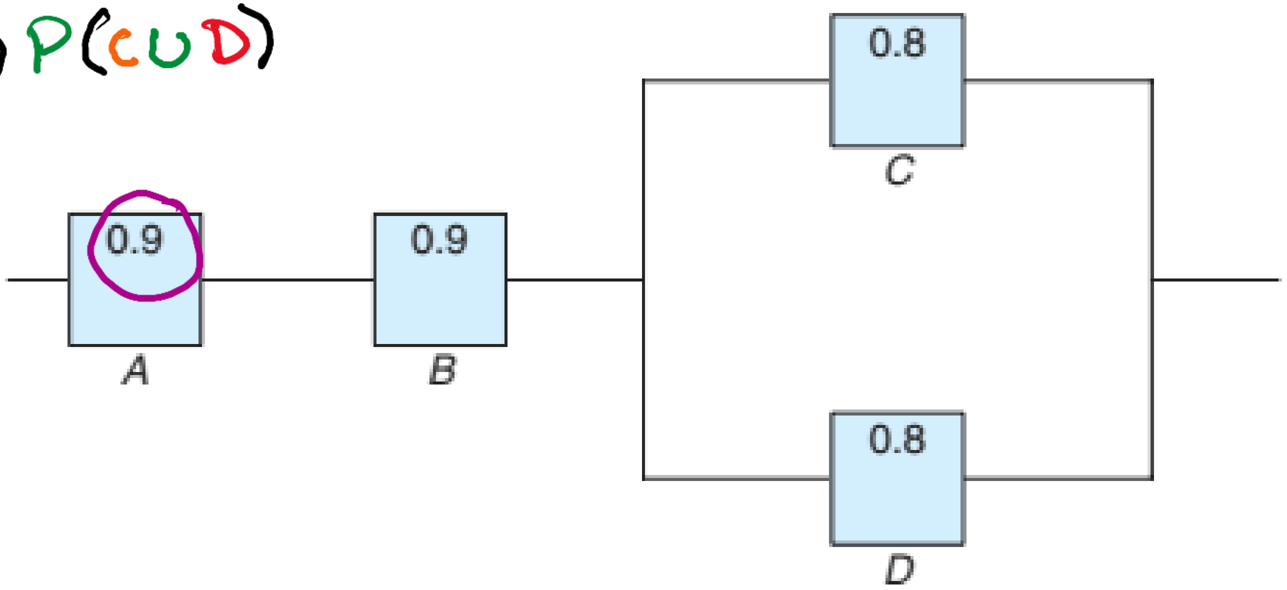
$$= P(A)P(B)[1 - P(C')P(D')]$$

$$= 0.9$$



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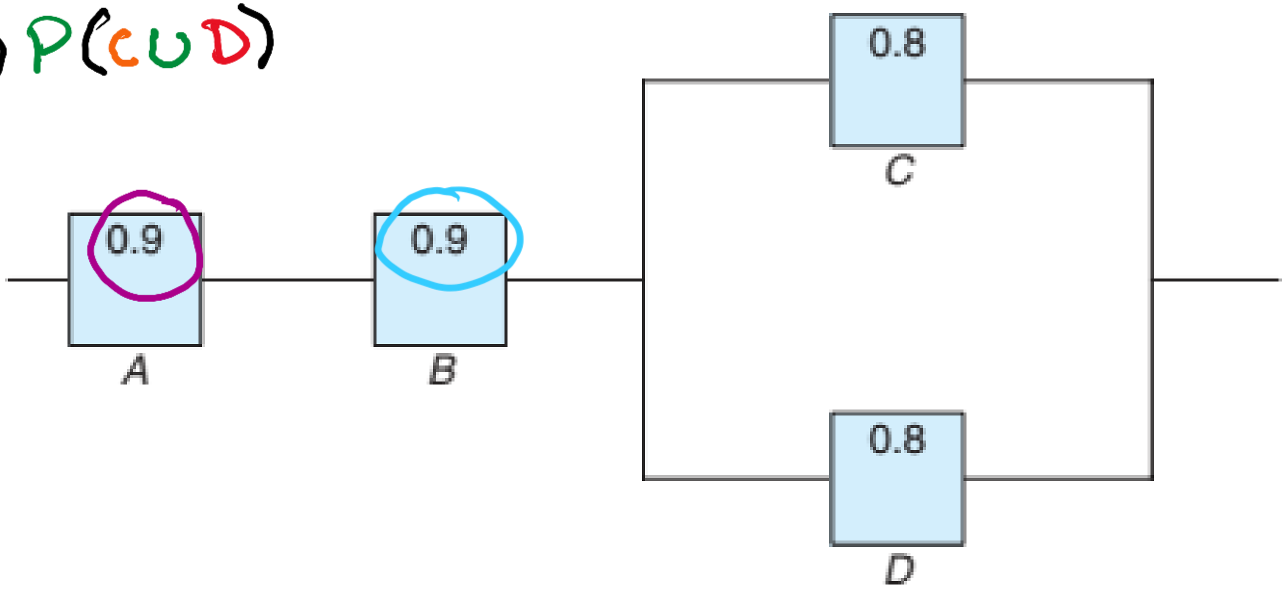
$$= P(A)\,P(B)\,[1 - P(C')\,P(D')]$$

$$= 0.9 \times 0.9$$



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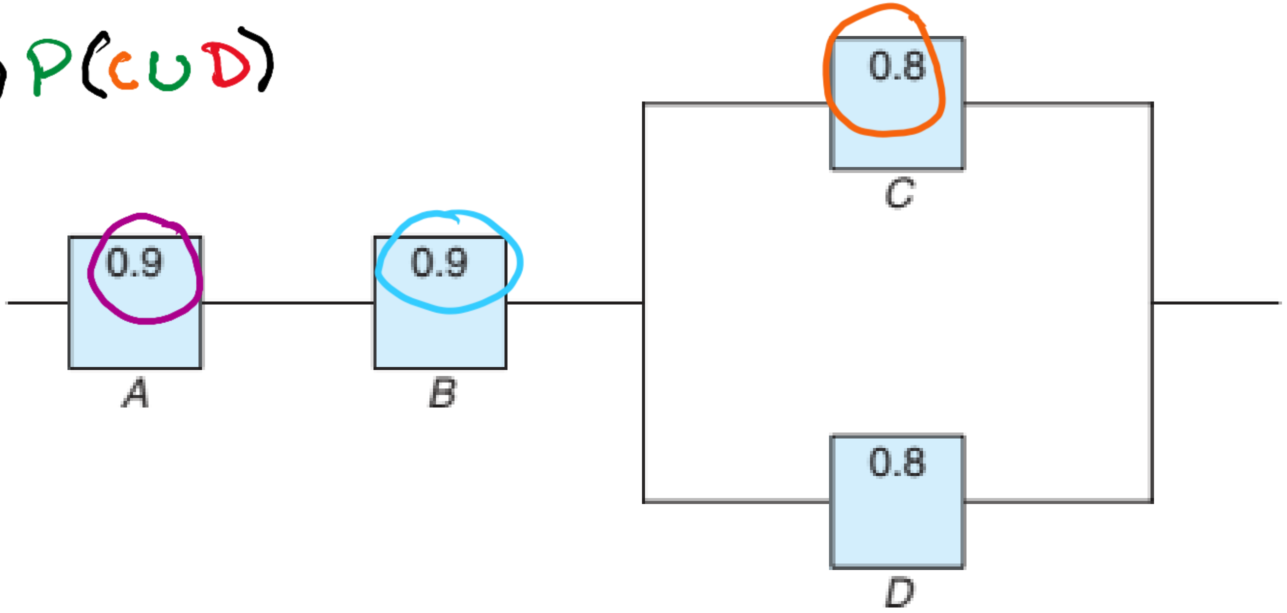
$$= P(A)\,P(B)\,[1-P(C')\,P(D')]$$

$$= 0.9\times 0.9\times[1-(1-0.8)$$



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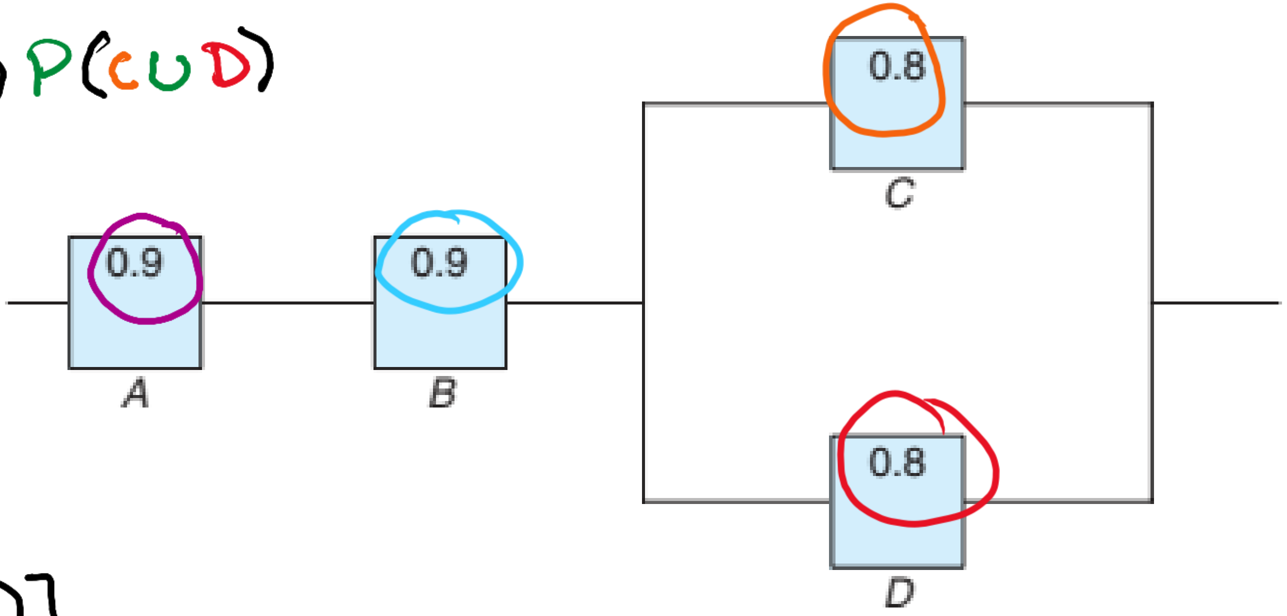

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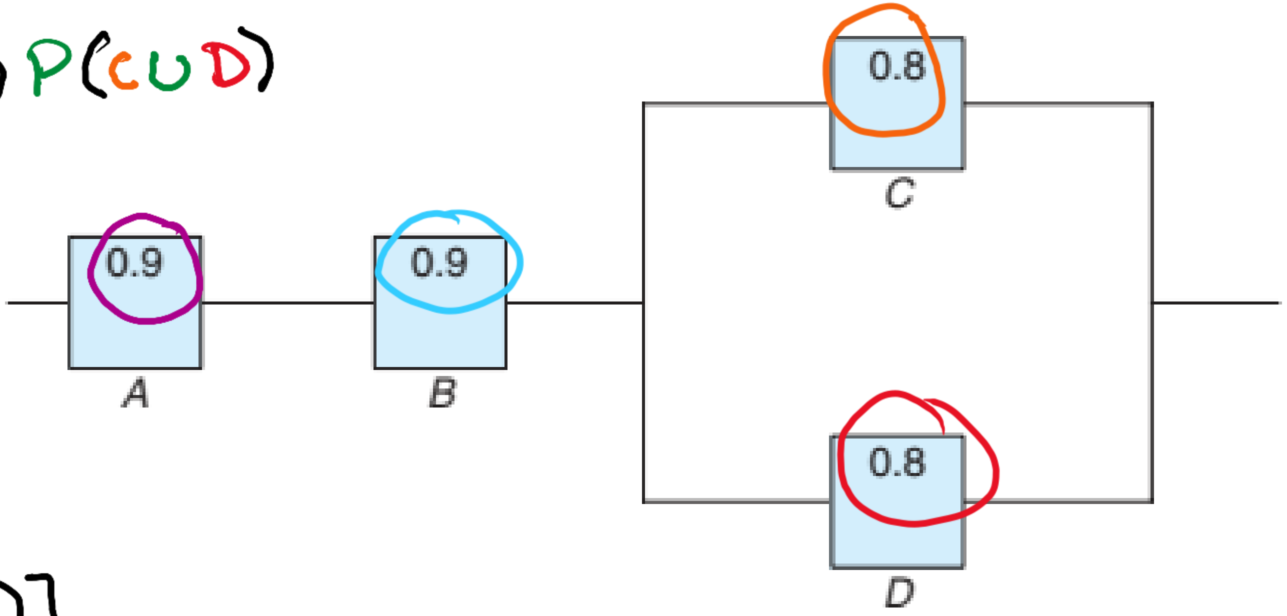
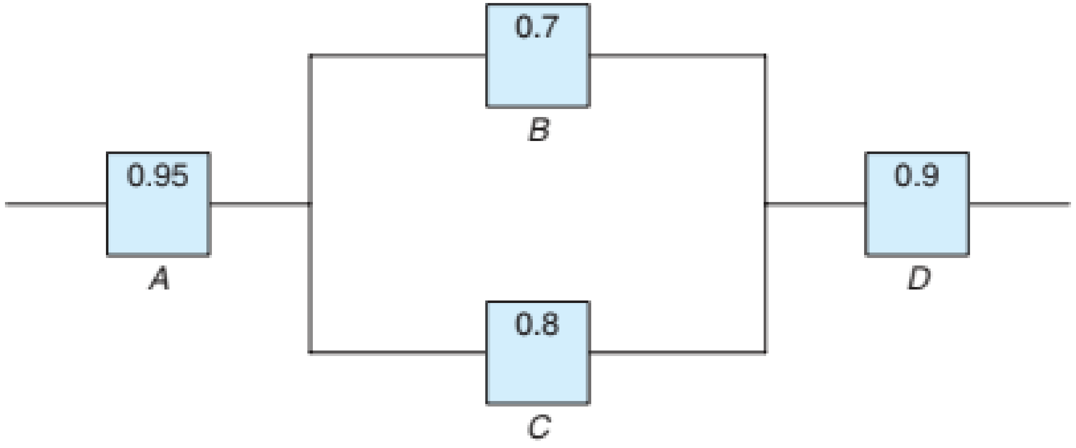
$$= 0.77$$ Ans



Suppose the diagram of an electrical system is as given in Figure. **What is the probability that the system works**? **Assume the components fail independently.**



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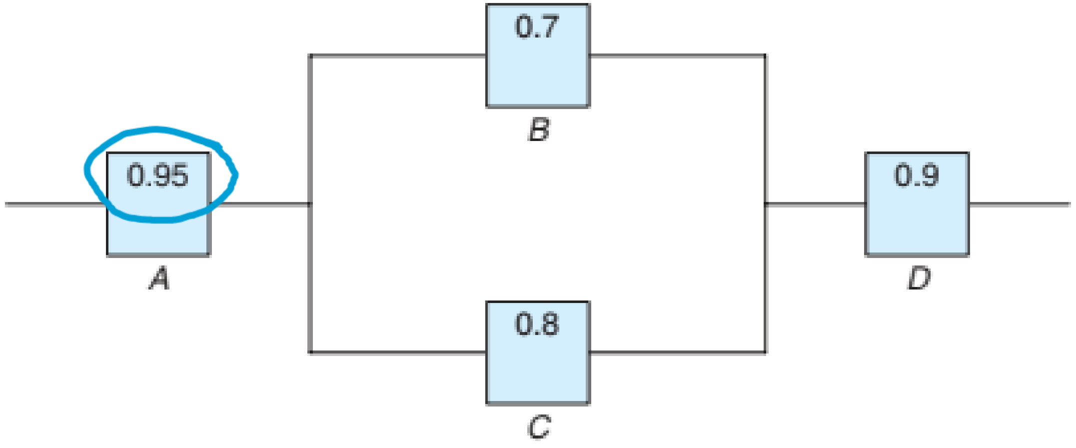
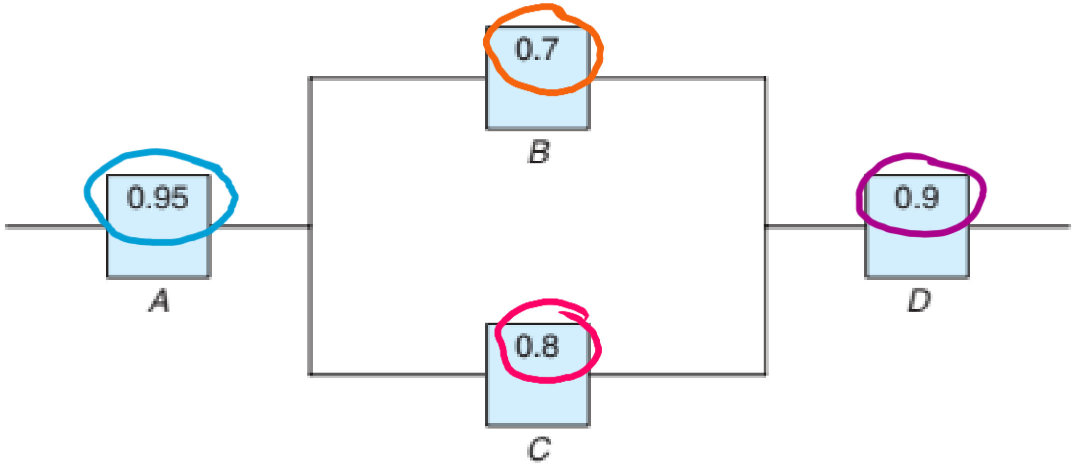


$P(A) = 0.95$

Suppose the diagram of an electrical system is as given in Figure. **What is the probability that the system works**? **Assume the components fail independently.**



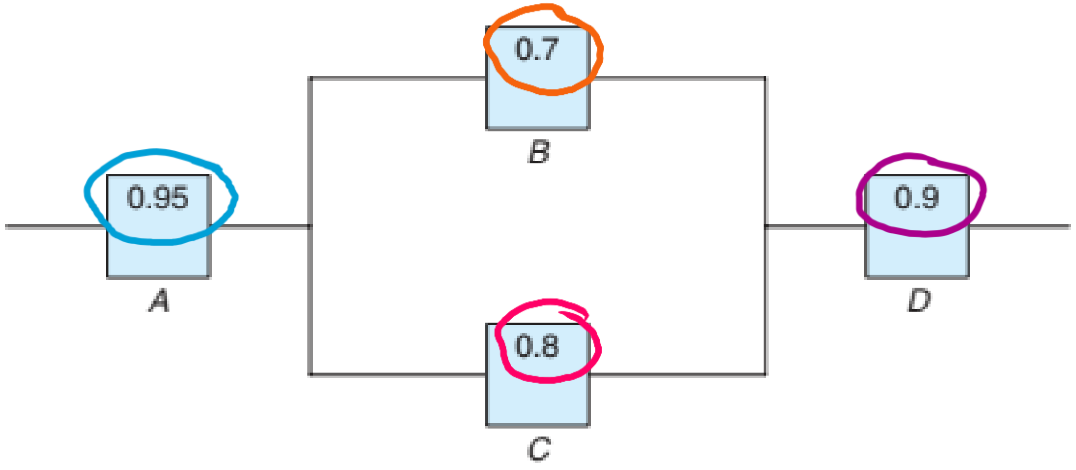
$P(A) = 0.95$

$P(B) = 0.7$

$P(C) = 0.8$

$P(D) = 0.9$

Suppose the diagram of an electrical system is as given in Figure. **What is the probability that the system works?** **Assume the components fail independently.**



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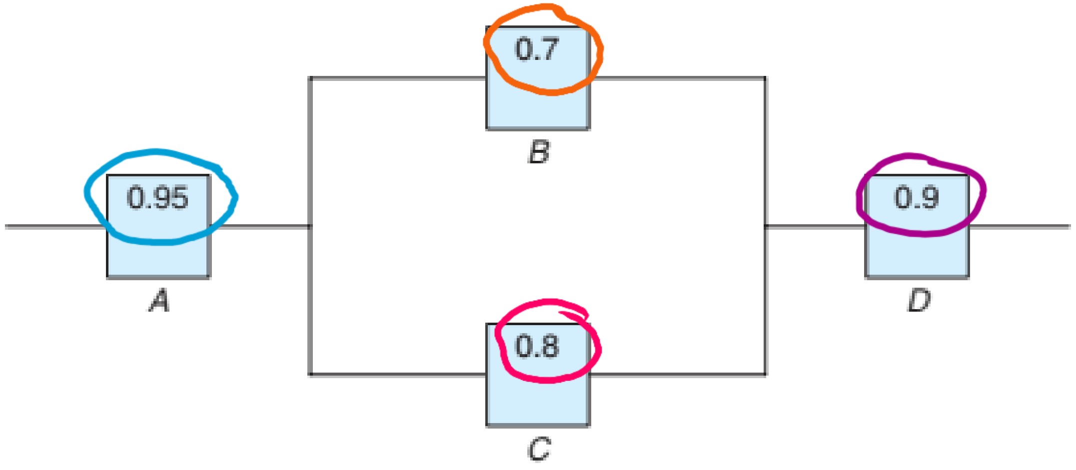
$P(A$



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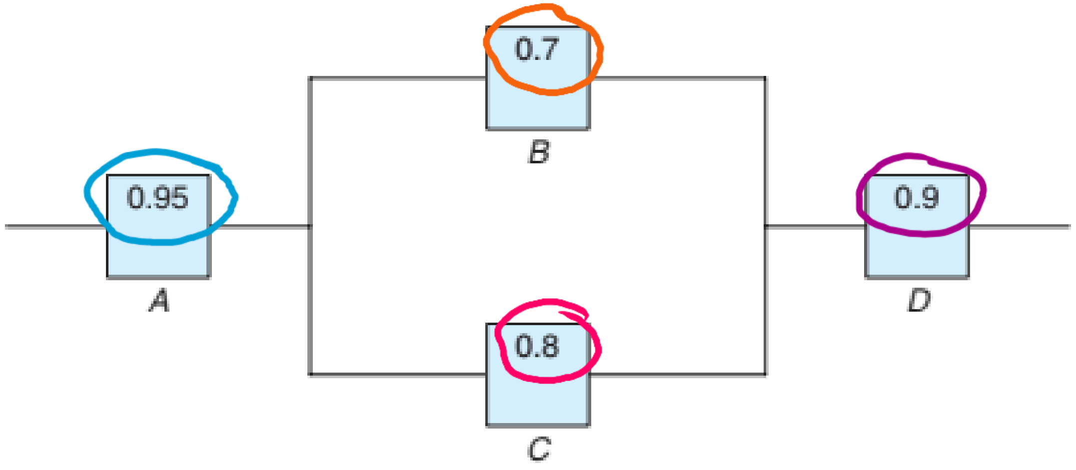
Suppose the diagram of an electrical system is as given in Figure. **What is the probability that the system works?** **Assume the components fail independently.**

$P(A \cap$



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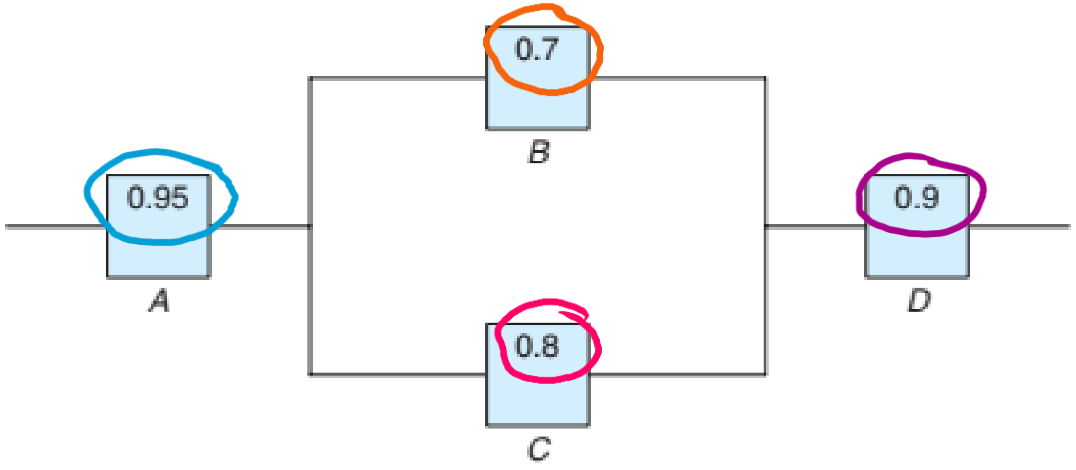
Suppose the diagram of an electrical system is as given in Figure. **What is the probability that the system works? Assume the components fail independently.**

$P(A \cap (B \cup C))$



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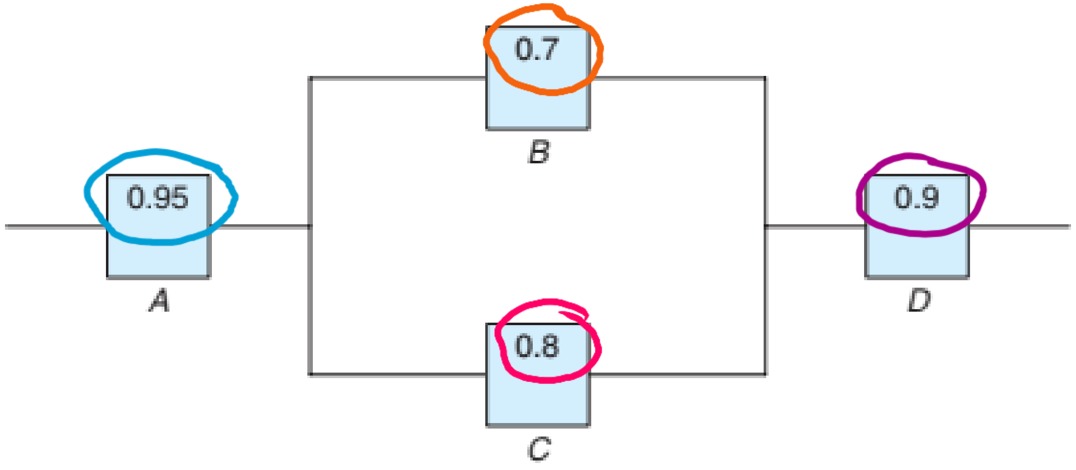
Suppose the diagram of an electrical system is as given in Figure. **What is the probability that the system works? Assume the components fail independently.**

$P(A \cap (B \cup C) \cap D) =$



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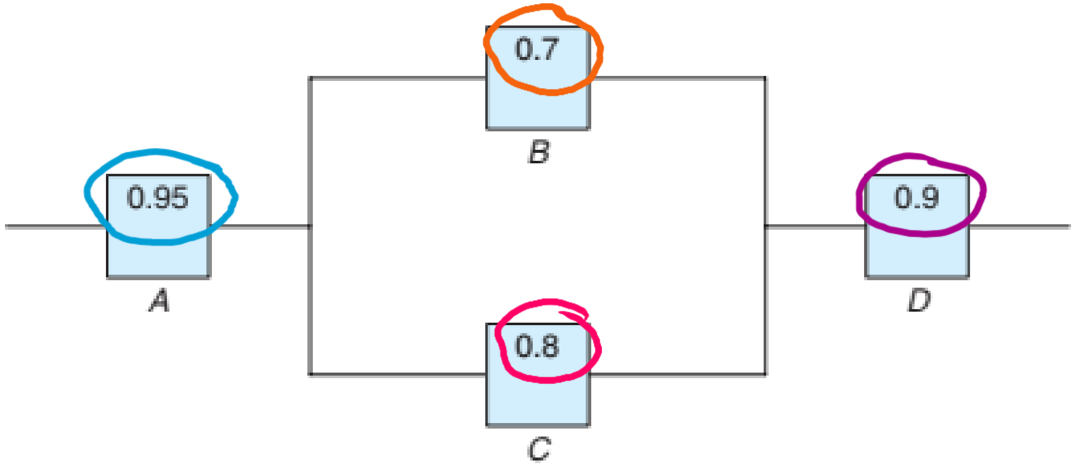

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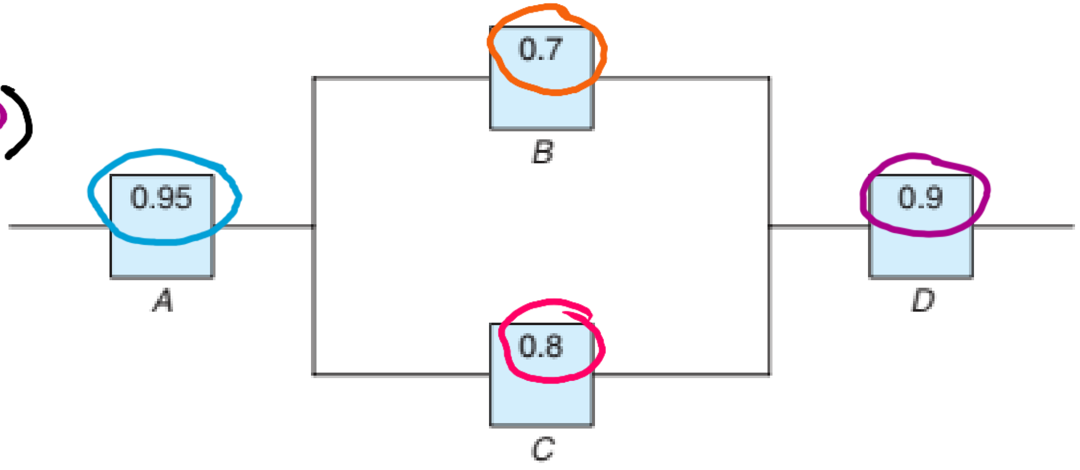
$$P(A \cap (B \cup C) \cap D) = P(A)\, P(B \cup C)\, P(D)$$



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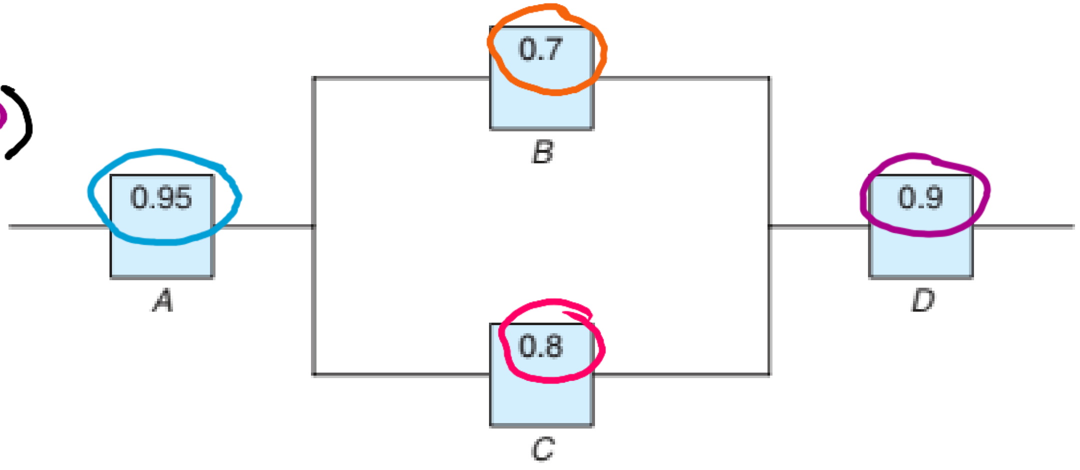

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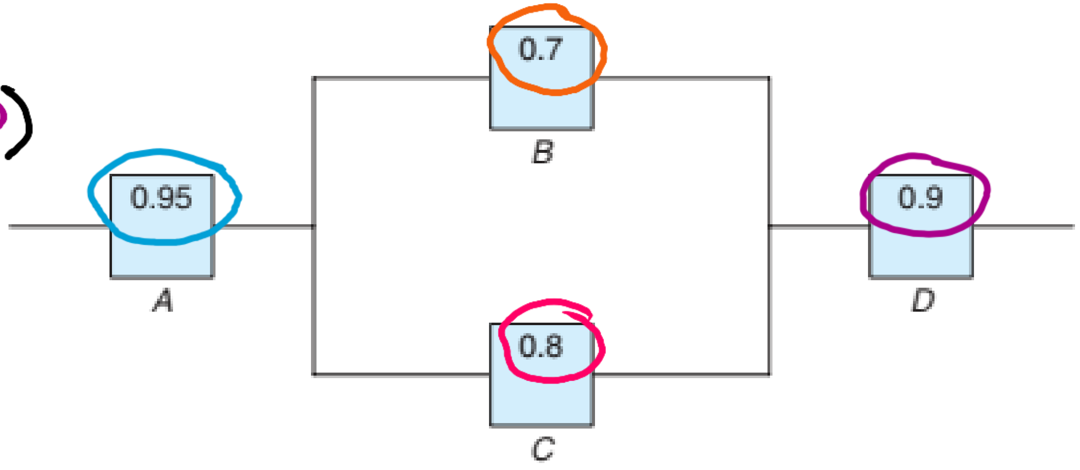

$$\boxed{P(A) = 0.95}$$

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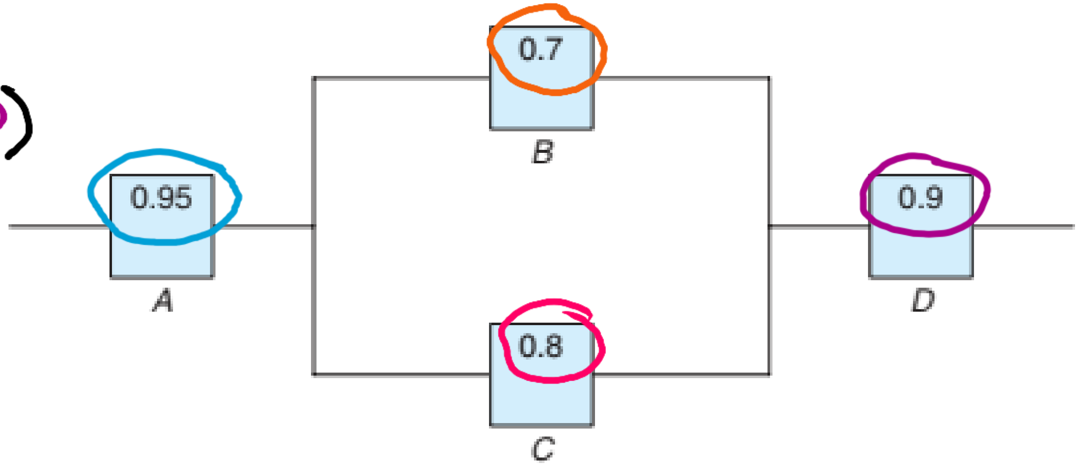

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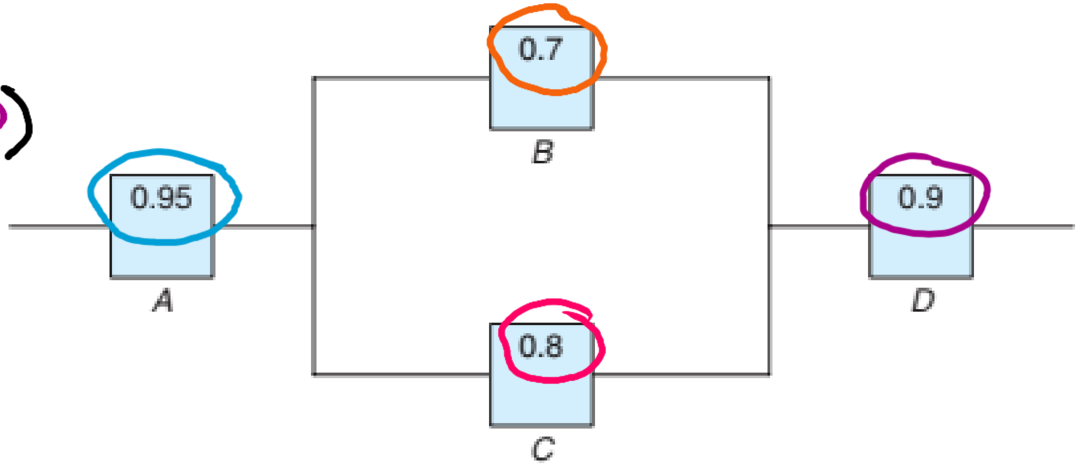

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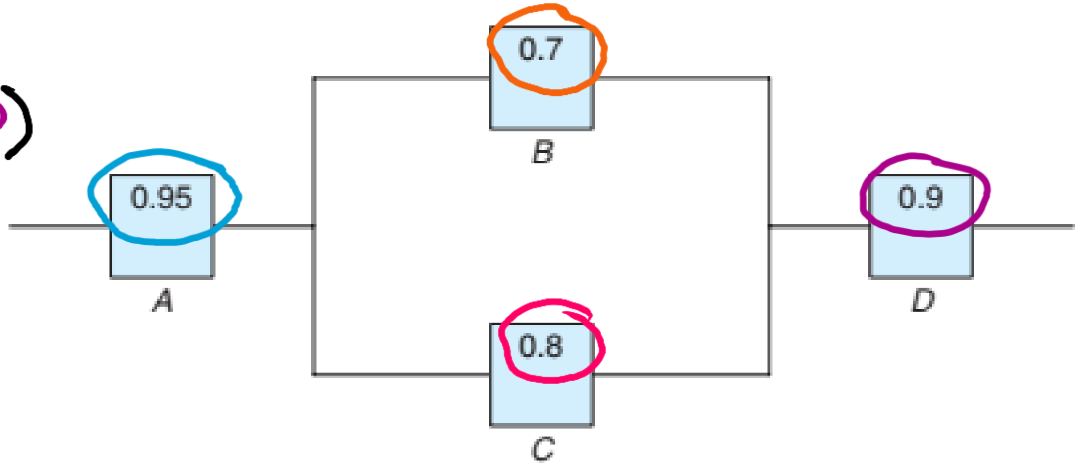

$$P(A \cap (B \cup C) \cap D) = P(A)\,P(B \cup C)\,P(D)$$

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$$= 0.95 \times 0.9 \times [\qquad\qquad]$$

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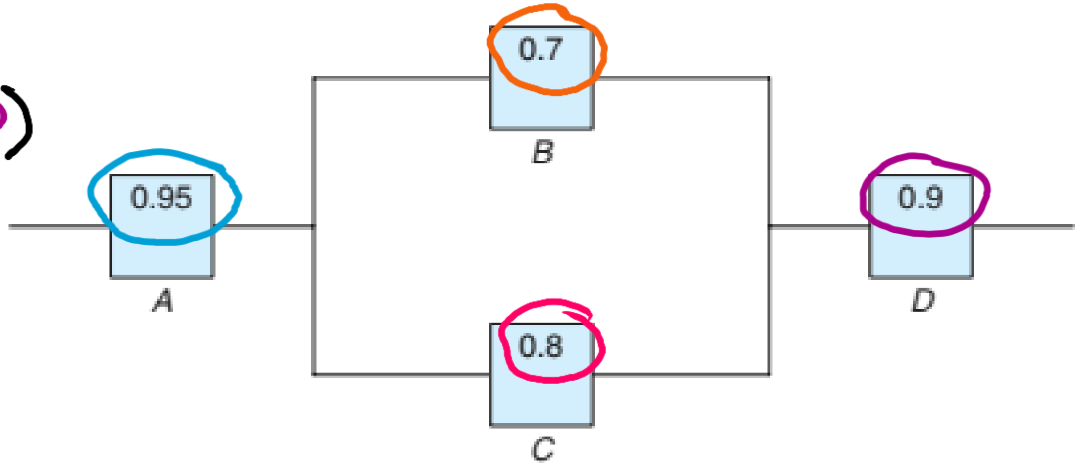

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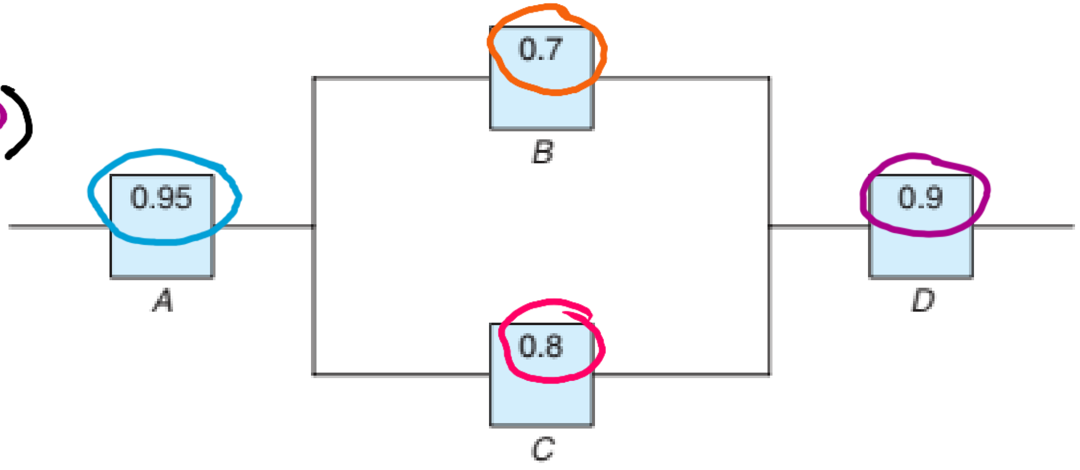

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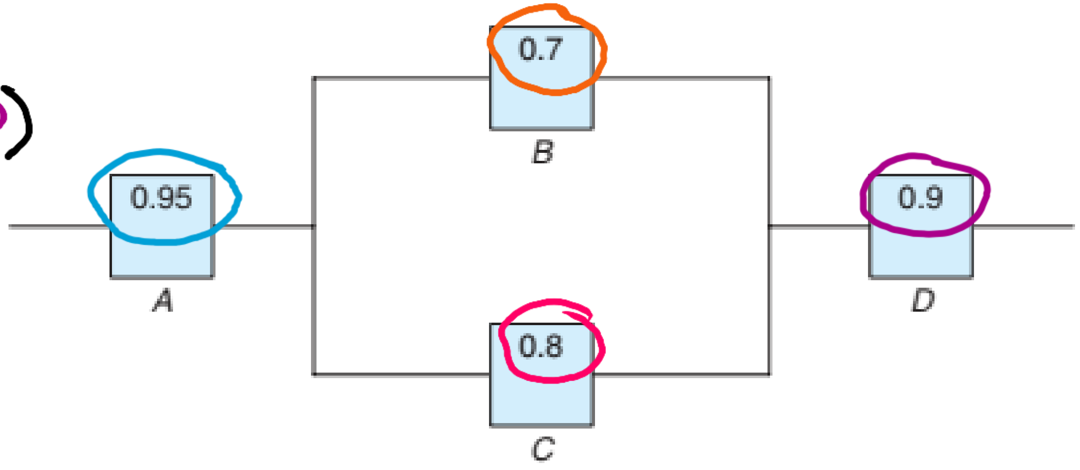

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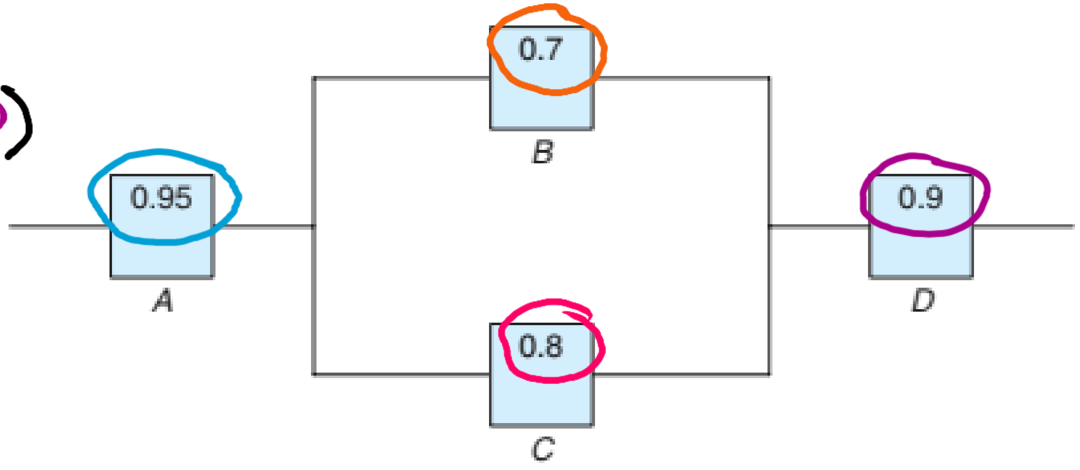

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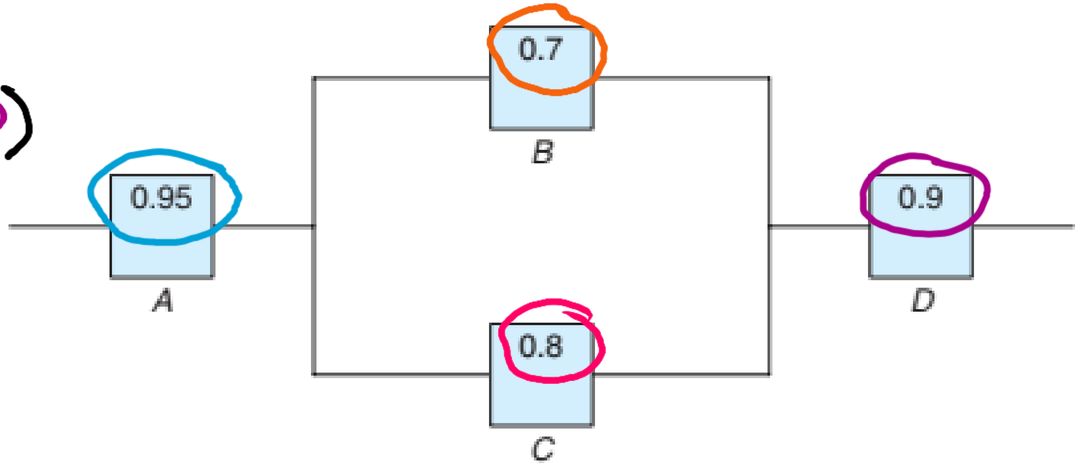

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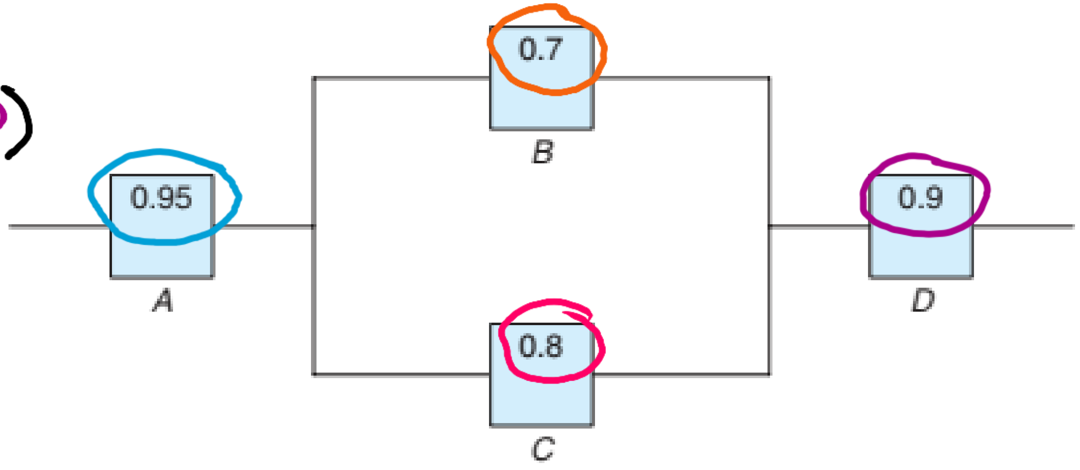

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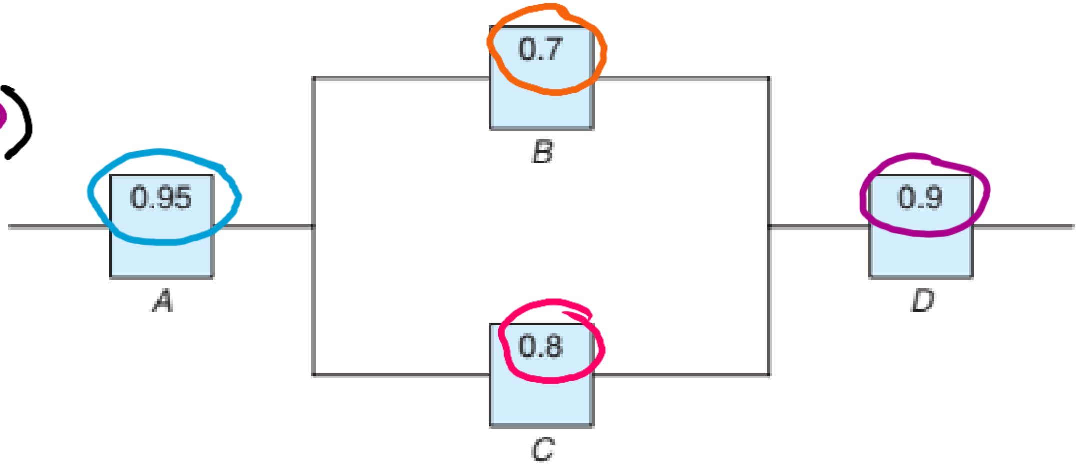

$$P(A \cap (B \cup C) \cap D) = P(A)\,P(B \cup C)\,P(D)$$

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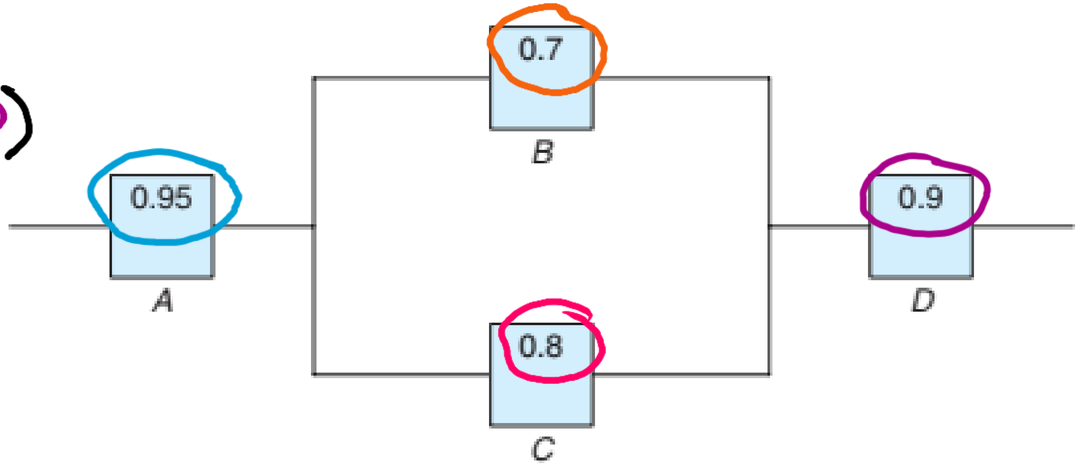

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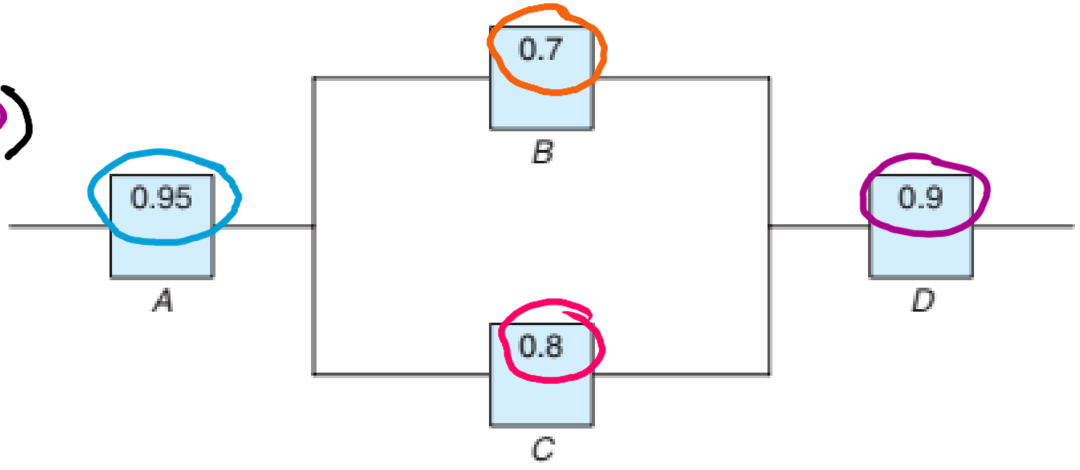

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$$= 0.95 \times 0.9 \times [1 - (1 - P(B))(1 - P(C))]$$

$$= 0.95 \times 0.9 \times [1 - (1 - \quad)(1 - P(C))]$$

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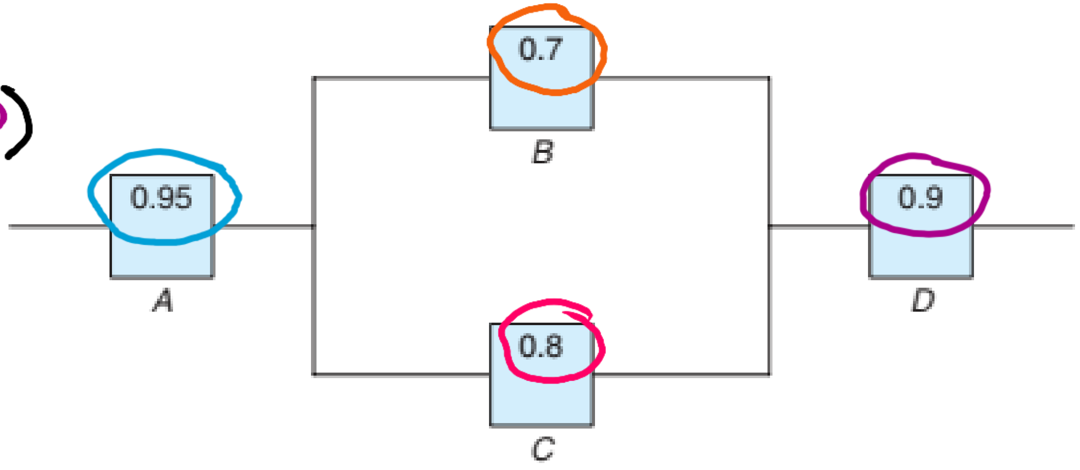

$$P(A \cap (B \cup C) \cap D) = P(A)\,P(B \cup C)\,P(D)$$

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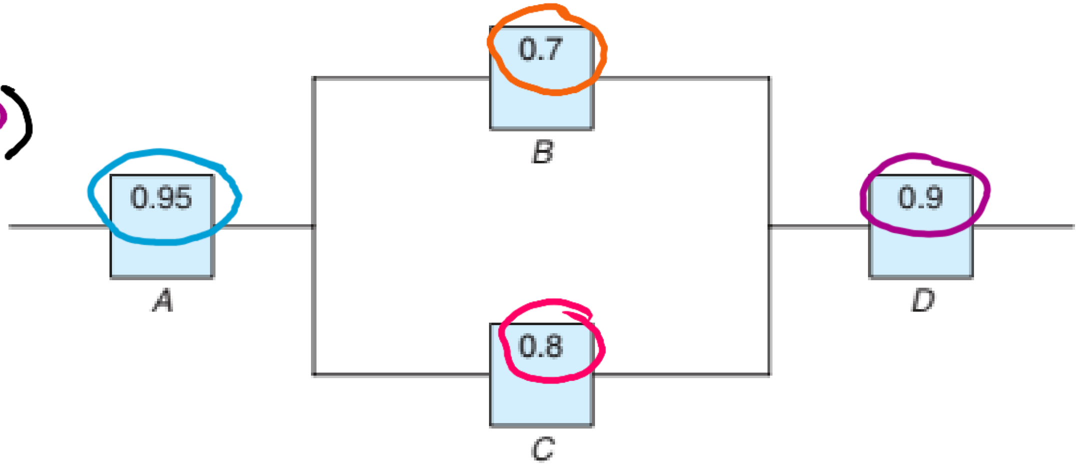

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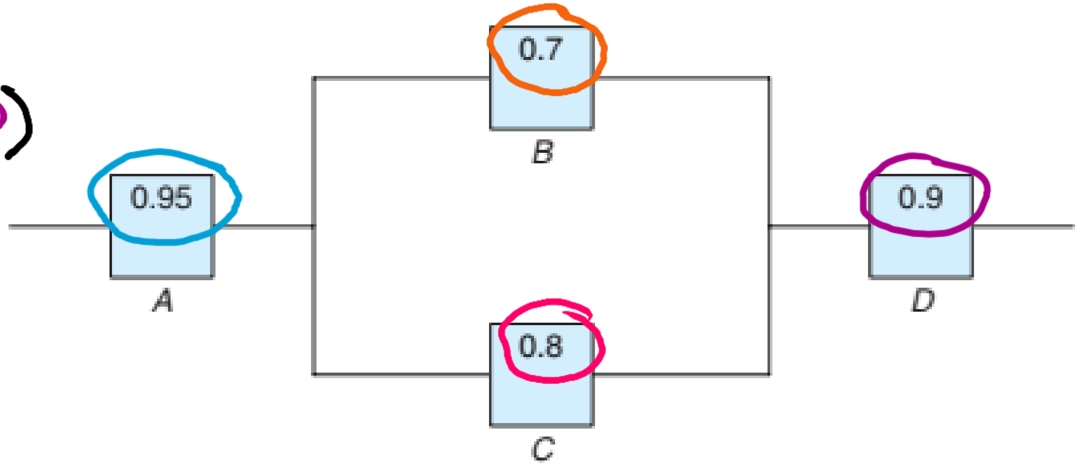

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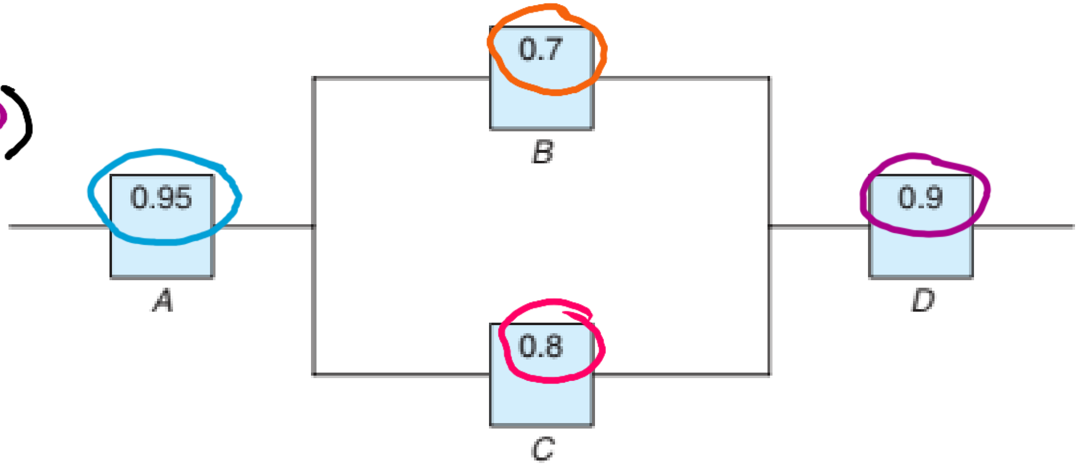

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$$= 0.8037 \text{ Ans}$$

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The probability that a doctor correctly diagnoses a particular illness is 0.7. Given that the doctor makes an incorrect diagnosis, the probability that the patient files a lawsuit is 0.9. What is the probability that the doctor makes an incorrect diagnosis and the patient sues?

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$$= 0.3 \times 0.9 = 0.27$$ Ans.

Pollution of the rivers in the United States has been a problem for many years. Consider the following events:

A: the river is polluted,

B : a sample of water tested detects pollution,

C : fishing is permitted

Assume

$P(A) = 0.3$, $P(B|A) = 0.75$, $P(B|A') = 0.20$,

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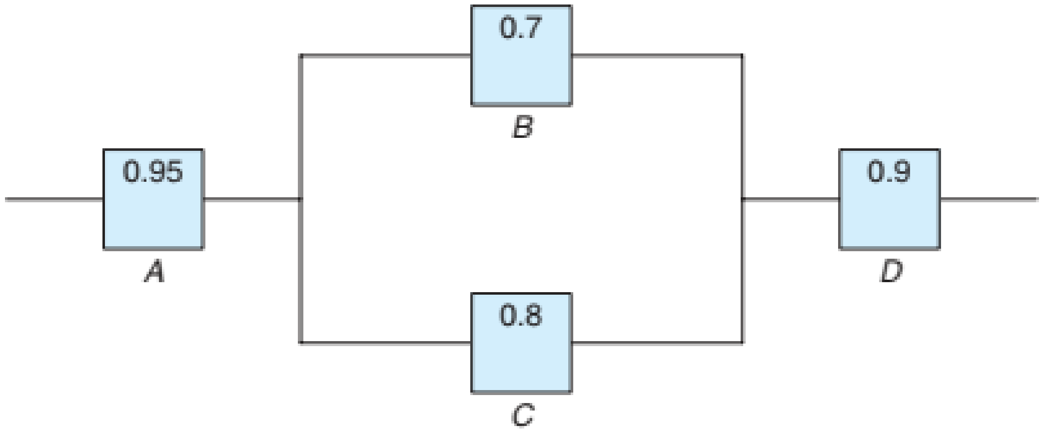

(a) Find $P(A \cap B \cap C)$.

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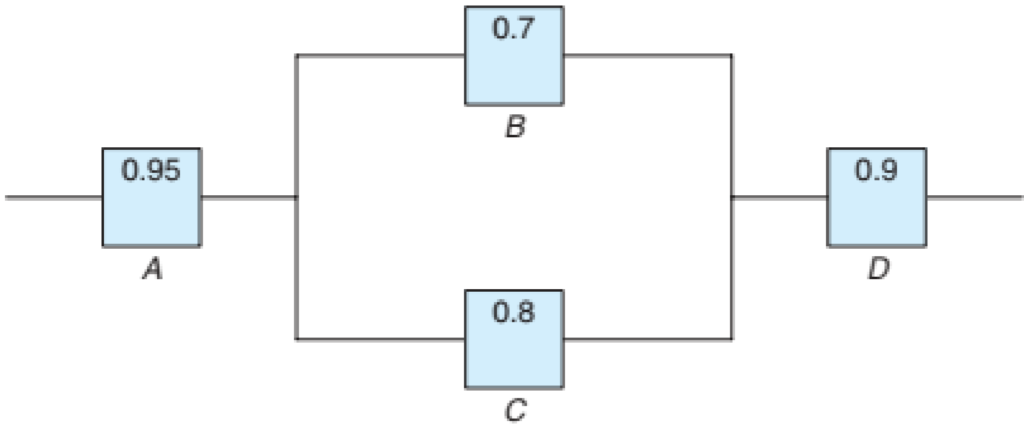

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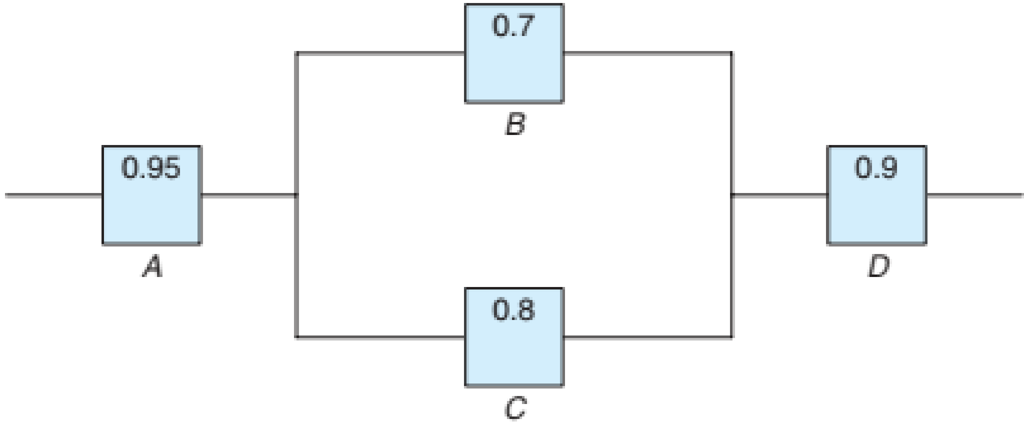

(a) Find $P(\underbrace{A \cap B}_{D} \cap C)$.

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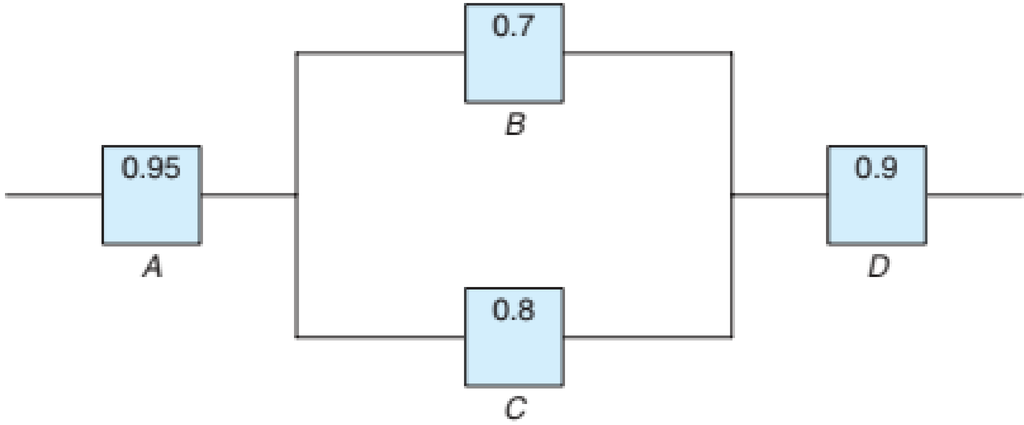

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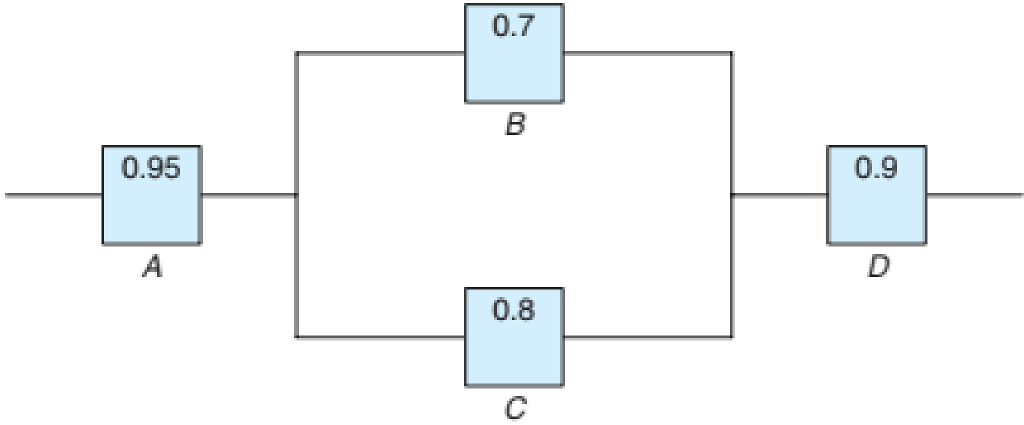

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D

$P(D \cap C) =$ $P(C|D)$

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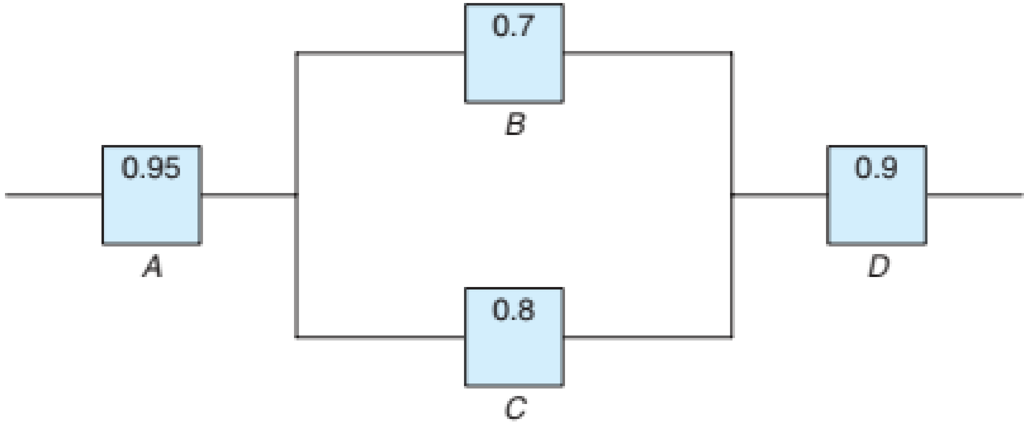

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$P(D \cap C) = P(D) \times P(C|D)$

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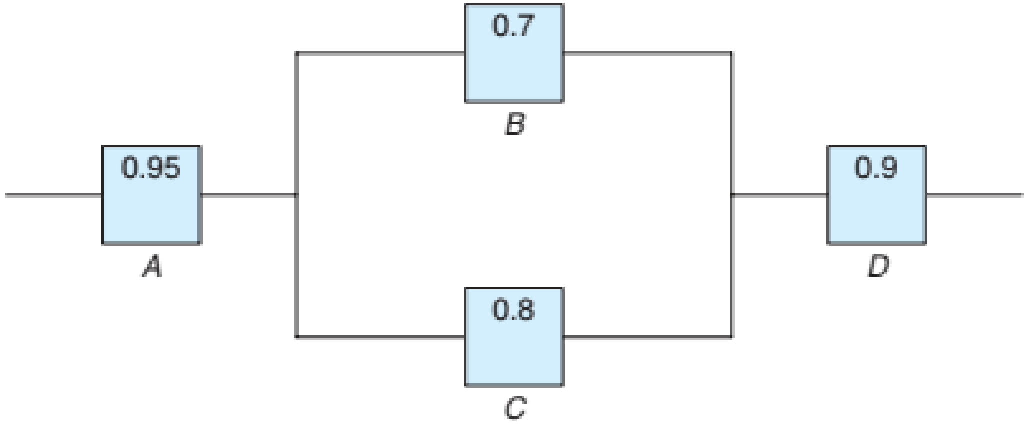

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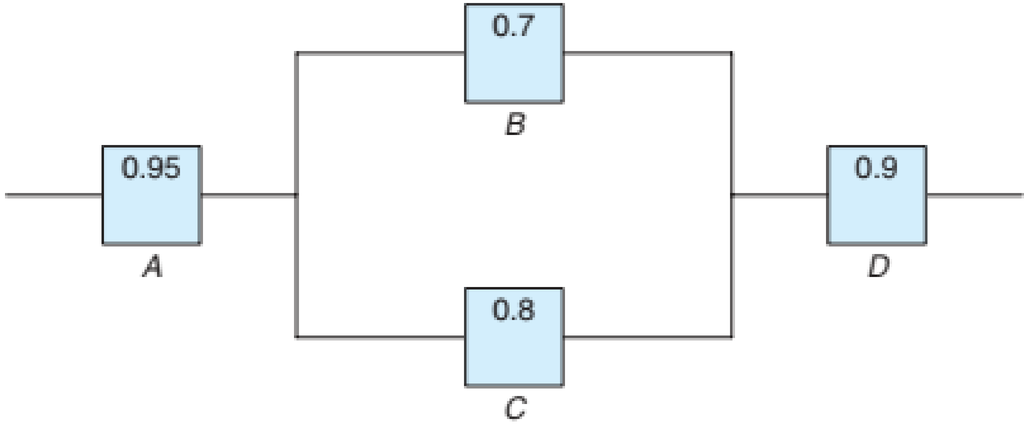

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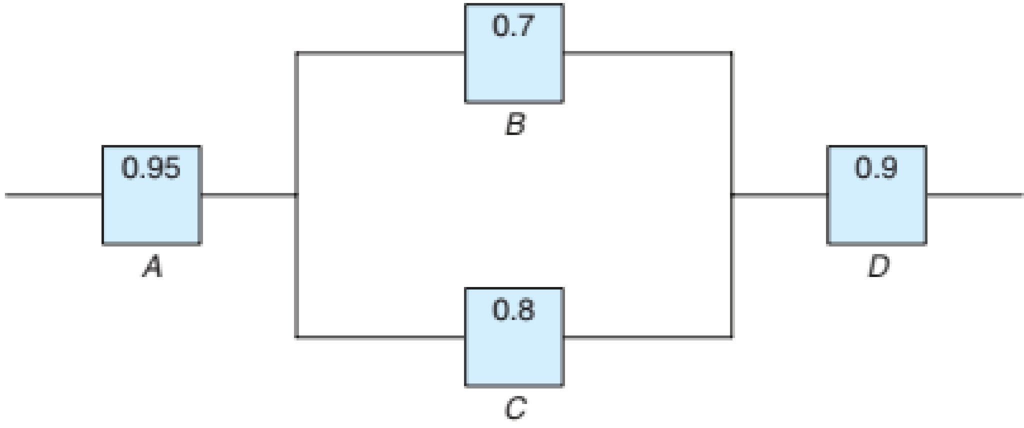

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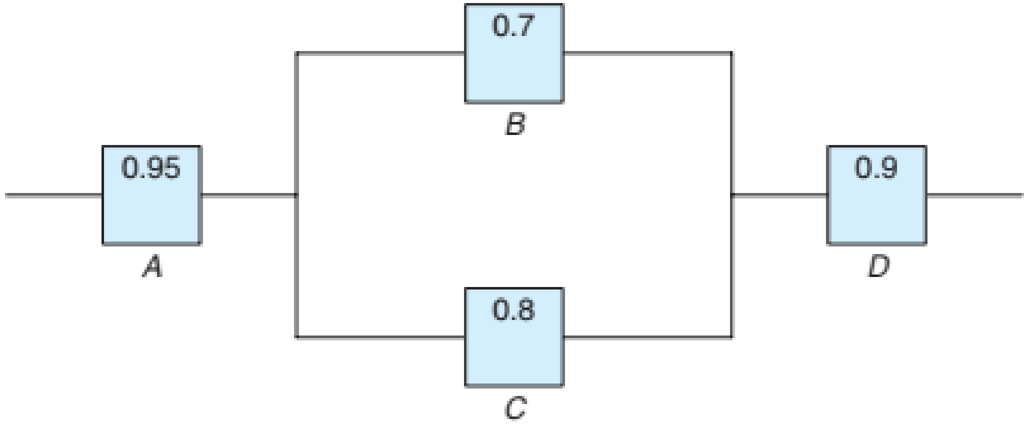

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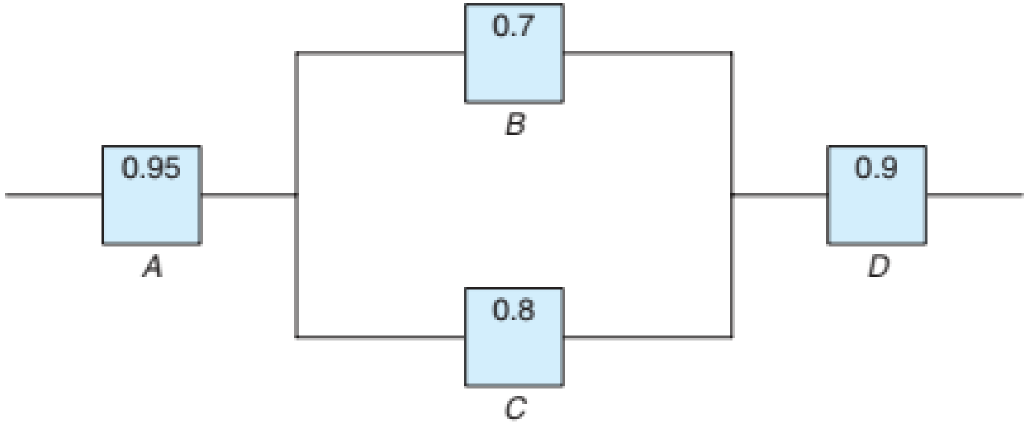

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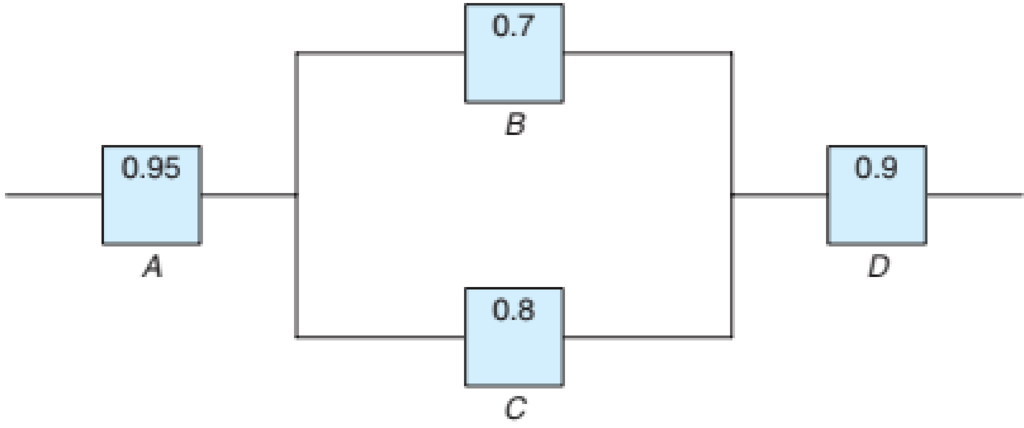

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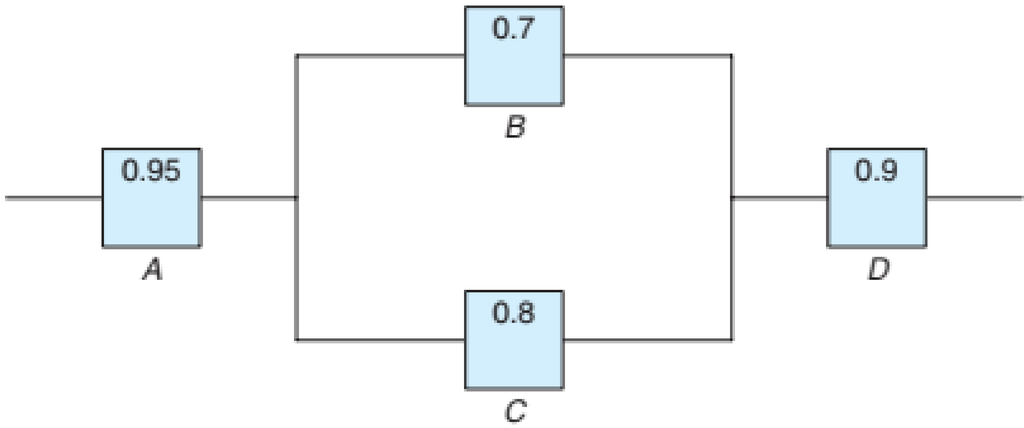

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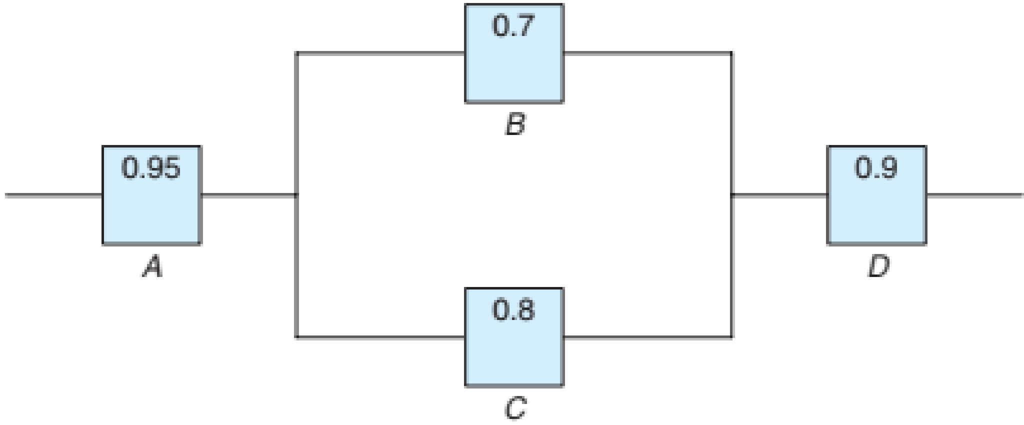

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$$= 0.3 \times 0.75 \times 0.20$$

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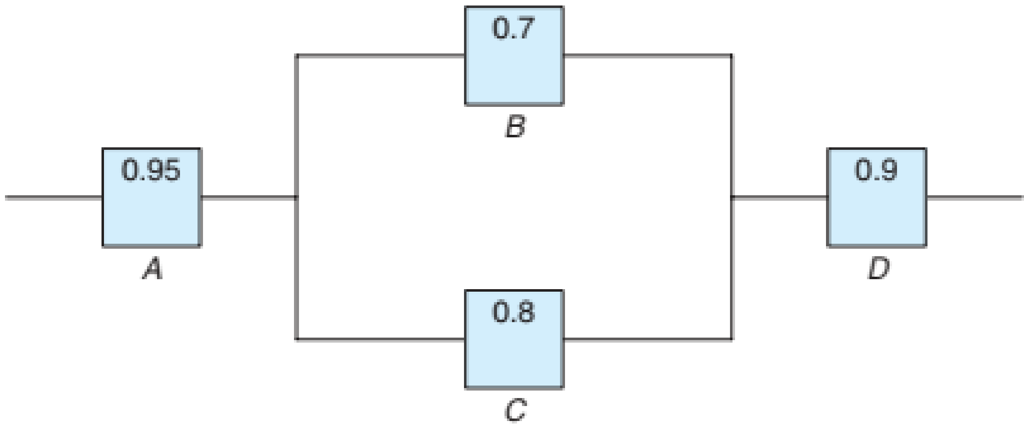

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Ans